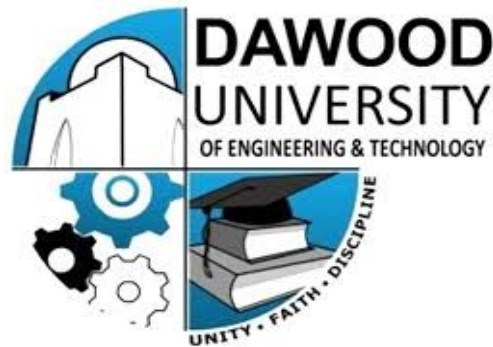


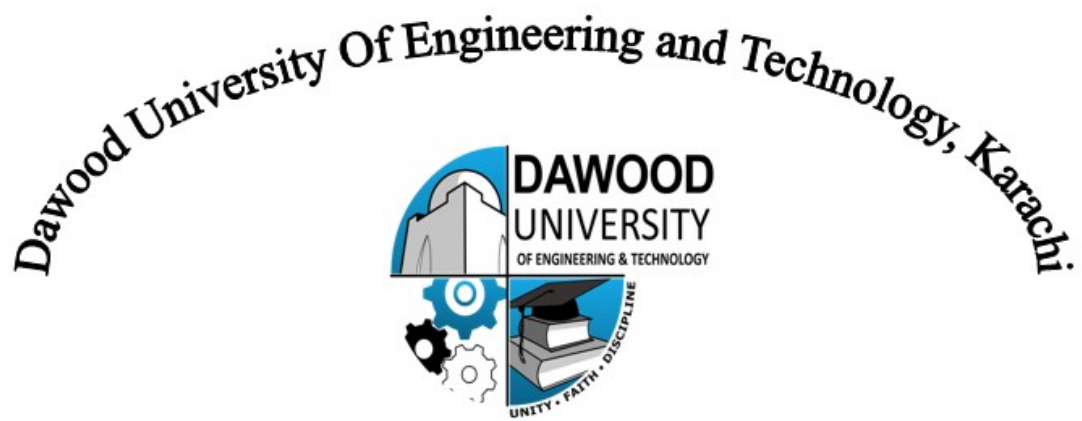
Introduction to Information and Communication Technologies
(Practical Manual)



Department of Artificial Intelligence

1st Semester-2025/F batch

Instructor Name: Engr. M. Ahsan Riaz



CERTIFICATE

This is to certify that Mr./

Ms. _____ bearing Roll# _____ of Batch 2025/

Has successfully completed all the labs prescribed for the course "Introduction to Information and Communication Technologies".

Lab(Instructor)

External Examiner

CHAIRPERSON

Department Of Artificial Intelligence

Dawood University of Engineering and Technology, Karachi, Pakistan

DATED: _____

1.CourseLearningOutcomes

MappingofCLOsandPLOs			
S.No.	CourseLearningoutcomes	PLOs	BloomsTaxonomy
1	Practice theICTexperimentsaccordingtothegiveninstructions.		P-3
2	Manipulate withguidancetheC-languageprogramsforvariousapplications.		P-3
3	Assume responsibilitytofollowlabrules,maintainlabmanualandbehaveethically.		A-3

AssessmentPlan

Assessmenttools	CLO-1	CLO-2	CLO-3	Total
OpenEndedLab-I	10marks			10
OpenEndedlab-II		10Marks		10
LabReport			10Marks	10
Practical	10Marks			10
VIVA	10Marks			10
Total	30Marks	10Marks	10Marks	50

DawoodUniversityofEngineering&Technology,Karachi

DepartmentofElectronicEngineering

RUBRICFOREVALUATIONOFSESSIONAL(OpenEndedlab-I)

Subject:Computerfundamental&Programming(Practical)

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Development of Component	Less than 40% (0.5)	70% (2)	All components Developed (3)
Implementation/Format %	40% (0.5)	80% (2)	100% results (4)
Error (faulty data) Missing important information	Major error (0.5)	Develop with few errors (2)	Error free (3)

Dawood University of Engineering & Technology, Karachi
Department of Electronic Engineering

RUBRIC FOR EVALUATION OF SESSIONAL (Open Ended lab-II)

Coding on C		
Coding	Develop but with few Errors	Without Errors
	1-3	4-7

CodeSimulation	Partiallycomplete 1-3	Completedsimulation 4-7
Results	Partiallyresultsobtain 1-3	Completed 4-6

RUBRICFOREVALUATIONOFFINAL

Subject:Computerfundamental&Programming(Practical)

DevelopmentofComponent	Lessthan40% (0.5)	70%. (2)	AllcomponentsDevelop (3)
Implementation/Format %	40% (0.5)	80% (2)	100%results (4)
Error(faultydata) Missingimportantinfor	Majorerror (0.5)	Developwithfewer rrors	Errorfree (3)

mation		(2)	
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DawoodUniversityofEngineering&Technology,Karachi
DepartmentofElectronicEngineering

RUBRICFOREVALUATIONOFVIVA

Subject:Computerfundamental&Programming(Practical)

VIVA			
Criteria	Improvementrequired(1-2)	Adequate(3-4)	Satisfactory(5)
Confidence	PoorBodylanguage	StableEyecontact&decentbodylanguage	ProperEyetoeyeccontact&Appropriatebodylanguage
Knowledge	PartiallycorrectTechnicalanswers.	LeastDescriptiveAnswers	AppropriatelevelofDescriptionofTechnicalAnswers.
Communication	Poor(difficulttospeak	Good(speakwithpartialfluency)	Excellentfluency
Appearance	Inappropriate	Neat&clean	Goodappearance

Table of Content Lab Manual

S.No	Experiment	CLO	Level	Date of conduct	SIGNATURE
1	Introduction to computer system. Introduction of Fundamental hardware and peripheral inside the CPU.	CLO1	C1		
2	Study and familiarize the basic characteristics of hardware, peripheral and component of computer system & assembling and disassembling of computer hardware.	CLO1	C1		
3	Study and familiarize the basic characteristics of software's.	CLO1	P4		
4	Study and familiarize the basic part of Browser, Internet Services provider and Online Services Providers, Function and Features of Browser, Search Engines, Some Common Services available on Internet.	CLO1	P4		
5	Introduction to MS-Word.	CLO1	P4		
6	Introduction to Excess Digital Library.	CLO1	P4		
7	Open Ended LAB-I	CLO1	P4		
8	Introduction to MS-Powerpoint.	CLO1	P4		
9	Introduction to MS-Excel.	CLO1	P4		
10	Introduction to C Language Programming.	CLO2	P4		
11	Arithmetic & logical Operation in C. Decision-making in the C Programming language	CLO2	P4		
12	Looping Construct in C language. Decision-making the switch case and conditional operator.	CLO2	P4		
13	Open Ended LAB-II	CLO2	P4		

Date: _____

Lab#01 INTRODUCTION AND IDENTIFICATION TO COMPONENTS OF A COMPUTER SYSTEM

OBJECTIVES:

- O1: To familiarize with the computer system and its basic functions.
O2: Familiarization with the hardware components and ports commonly found in a real PC.
O3: To carefully note and follow safety protocols for working in the Lab.

EQUIPMENT TO BE USED IN THE LAB

- Standard PC Core i-5.

THEORY:

A computer is a multi-purposed device and is a complex system consisting of both *hardware* and *software* components.



Fig.1.1 Hardware and software

Hardware means the physical parts of the computer. The **hardware** components of a computer system are the electronic and mechanical parts.

Software means the machine-readable instructions (programs, data and operating information) used with the physical computer. The **software** components of a computer system are the data and the computer programs. It is information stored on computer hardware.

HARDWARE COMPONENTS

The major hardware components of a computer system are:

- Processor
- Memory
- I/O devices

Processor:

The **processor** is an electronic device about a one-inch square, covered in plastic. Inside the square is an even smaller square of silicon containing millions of tiny electrical parts. A modern processor may contain billions of transistors. It does the fundamental computing within the system, and directly or indirectly controls all the other components. The processor is sometimes called the **Central Processing Unit** or **CPU**. A particular computer will have a particular type of processor, such as a Pentium processor etc.

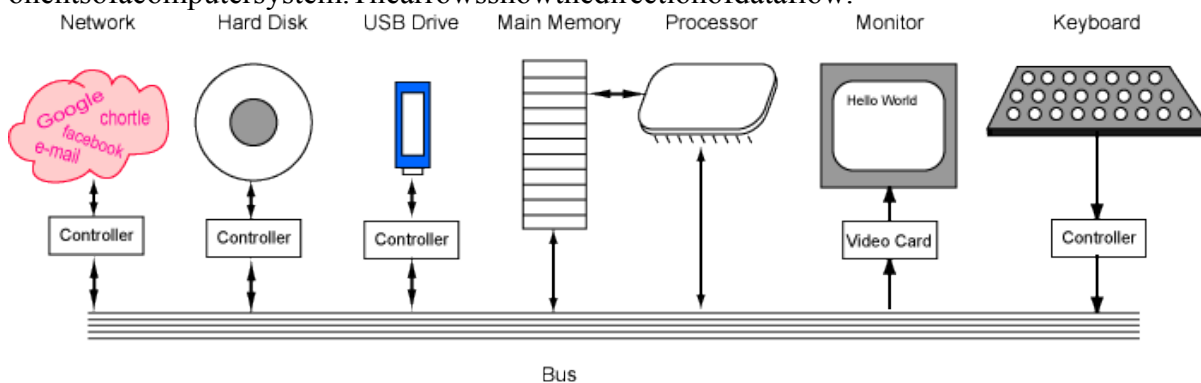
Memory:

A processor chip has relatively little memory. It has only enough memory to hold a few instructions of a program and the data they process. Complete programs and data sets are held in memory external to the processor. This memory is of two fundamental types: main memory, and secondary memory.

Main memory is sometimes called **volatile** because it loses its information when power is removed. **Secondary memory** is usually nonvolatile because it retains its information when power is removed. (However, it needs power when information is stored into memory or retrieved from it.) Main memory is sometimes called **main storage** and secondary memory is sometimes called **secondary storage** or **mass storage**.

I/O Devices:

The terms *input* and *output* say if data flow into or out of the computer. The picture shows the major hardware components of a computer system. The arrows show the direction of data flow.



Main Components of a Computer System

Fig. 1.2: Hardware Components

Input device is a peripheral (piece of computer hardware equipment) used to provide data and control signals to an information processing system such as a computer or other information appliance. Examples of input devices include keyboards, mouse, scanners, digital cameras and joysticks.

An **output device** is any piece of computer hardware equipment used to communicate the results of data processing carried out by an information processing system (such as a computer) which converts the electronically generated information into human-readable form. (monitor, speaker, headphones, Printer).

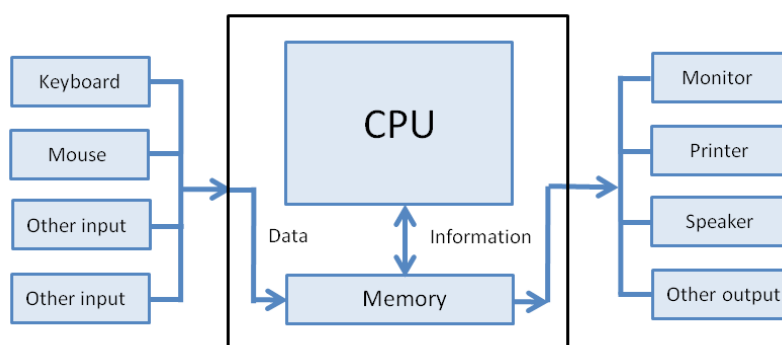


Fig. 1.3: The flow of information

SOFTWARE

Based on the goal, the computer software can be divided into two types: i) **Application Software** and ii) **System Software**

Application software, which uses the computer system to perform special functions or provide entertainment functions beyond the basic operation of the computer itself. There are many different types of application software, because the range of tasks that can be performed with a modern computer is so large - see list of software.

System software, which is designed to directly operate the computer hardware, to provide basic functionality needed by users and other software, and to provide a platform for running application software. System software includes:

- Operating systems, which are essential collection of software that manages resources and provides common services for other software that runs "on top" of them. Supervisory programs, boot loaders, shells and window systems are core parts of operating systems. In practice, an operating system comes bundled with additional software (including application software) so that a user can potentially do some work with a computer that only has an operating system.
- Device drivers, which operate or control a particular type of device that is attached to a computer. Each device needs at least one corresponding device driver; because a computer typically has at minimum at least one input device and at least one output device, a computer typically needs more than one device driver.

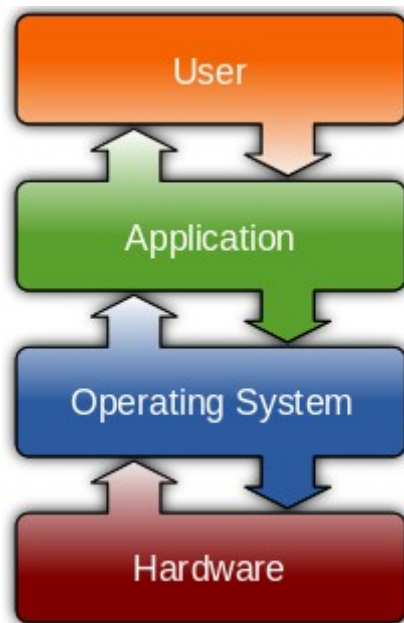


Fig1.4: A diagram showing how the operating system software and application software are layered on a typical desktop computer. The arrows indicate information flow.

Observations:

Component	Found(Yes/ No)	How many Items	Manufacturer Information	No.of Pins/ Holes
Power button				
Reset button				
Floppy drive				
CD/DVD drive				
Port for keyboard				
Port for mouse				
Serial port				

Parallelport				
LANport				
Powersocket				
USBport				
ModemPort				
GamePort				
Processorsocket				
Memoryslots				
PCIslots				
ISAslots				
BatteryforCMOS				
BIOSchip				
Motherboardchipset				
AGPslot				
Powersupply				
IDEControllerconnect ors				

LEARNING OUTCOMES

Upon successful completion of the lab, students will be able to:

LO1: Define & Differentiate between hardware and software.

LO2: Define & Differentiate between Input and output devices and their functions.

LO3: Define & Differentiate between application and system software.

CONCLUSION:

Date: _____

Lab#02 STUDY AND FAMILIARIZE THE BASIC CHARACTERISTICS OF HARDWARE, PERIPHERAL AND COMPONENTS OF COMPUTER SYSTEM & ASSEMBLING AND DISASSEMBLING OF COMPUTER HARDWARE.

OBJECTIVES:

- O1: To familiarize with the hardware components of computer system.
- O3: Familiarization with the often required operations like replacement and addition of RAM expansion cards.

EQUIPMENT TO BE USED IN THE LAB

- A computer system.
- RAM modules, a few PCI cards and other peripheral components.

THEORY:

Computer:

A computer is made up of integrated elements like CPU, input & output devices and storage devices. Each of these units performs a specific task. Computer is a device that takes input (Data), processes it and generates output (information).

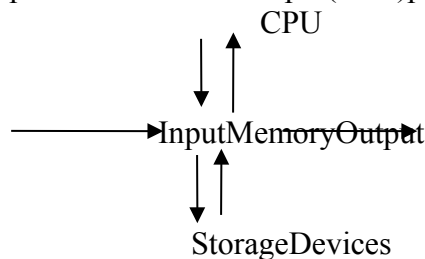


Fig.2.1: Flow of information

Hardware:

The physical attributes of the computer system. The hardware design specifies the commands it can follow, and the instructions tell it what to do. The hardware components included in the computer system are, Motherboard, Processor, Memory.

Motherboard:

It is the main circuit board; contains the connectors for attaching additional boards/cards. Typically, the motherboard contains the CPU, BIOS, memory, mass storage interfaces, serial and parallel ports, expansion slots, and all the controllers required to control standard peripheral devices, such as the display screen, keyboard, and disk drive. Collectively, all these chips that reside on the motherboard are known as the motherboard's chipset. To upgrade the PC, memory chip or CPU chip can be replaced.

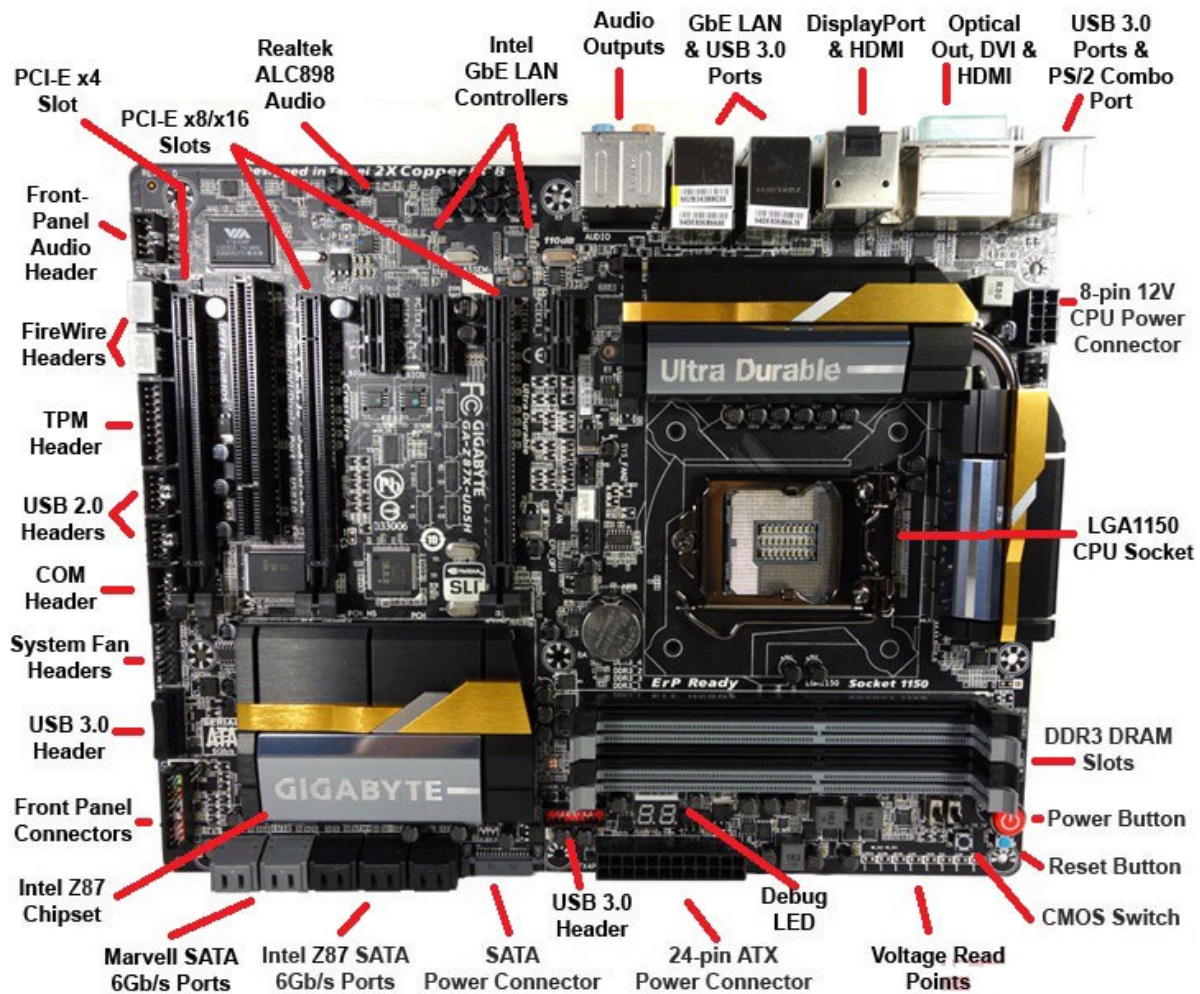


Fig.2.2: A typical motherboard

Processor Socket:

Several types of socket exist, but only 2 of them are really used, and both of them are used by Intel and AMD. The Pin Grid Array (PGA) and the Land Grid Array (LGA).

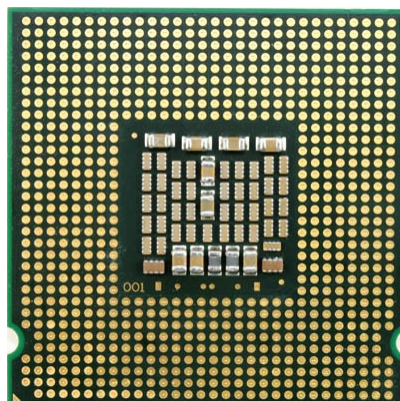


Fig.2.3:Socket775/SocketLGA775/SocketT

Memory Slots:

Memory slots also known as memory banks are for Random Access Memory modules (RAM). There are two of our memory banks: each memory bank can receive a RAM module designed for a specific Mother Board.

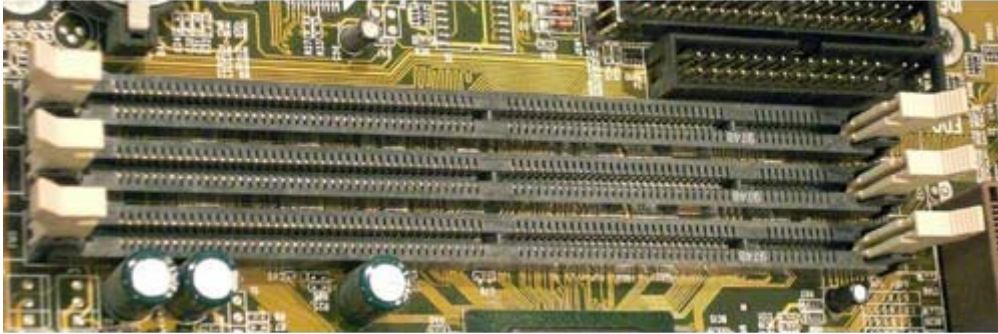


Fig.2.4:Computer Memory Slot

Processor:

Central Processing Unit, CPU, is the component installed in your motherboard socket. The CPU executes and interprets programs, and processes data.

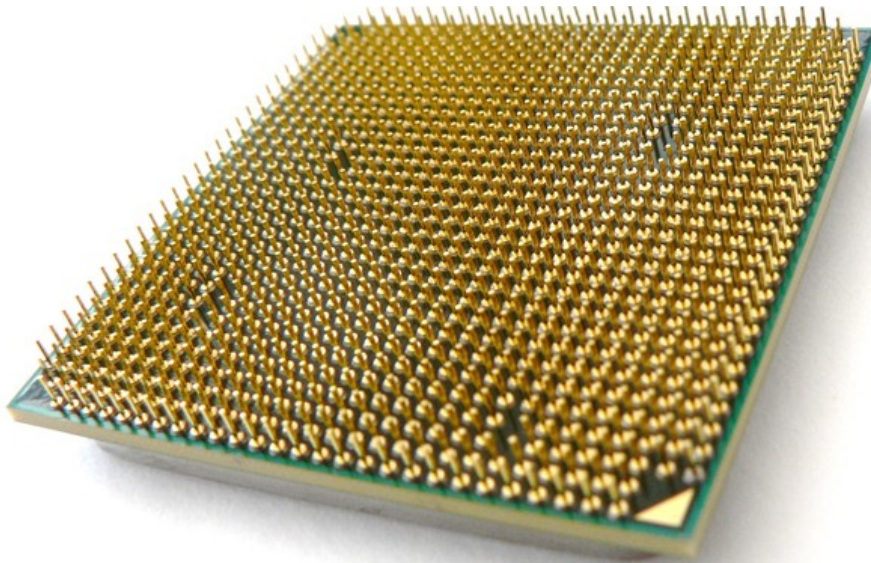


Fig2.5:Processor

Power Supply Connectors:

Main power connectors are plugged to the motherboard. The connector with 20 pins may be used on motherboards and the connector with 4 pins is used for extra power for CPU's and graphic cards. It is not necessary that 4 pins connector is used, it is used as a requirement.

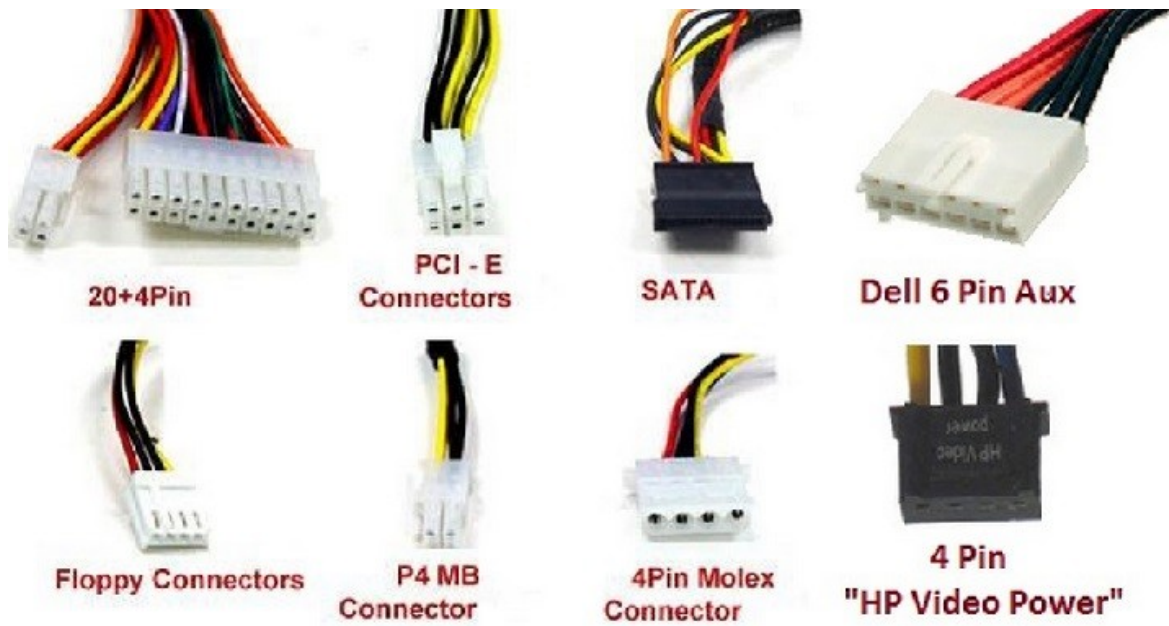


Fig.2.6:PowerSupplyConnectors

PCISlots,AGP andPCIe:

PeripheralComponentInterconnect(PCI)areusedtoinstallsoundcards,graphiccards,Ethernetcardsandmode ms.

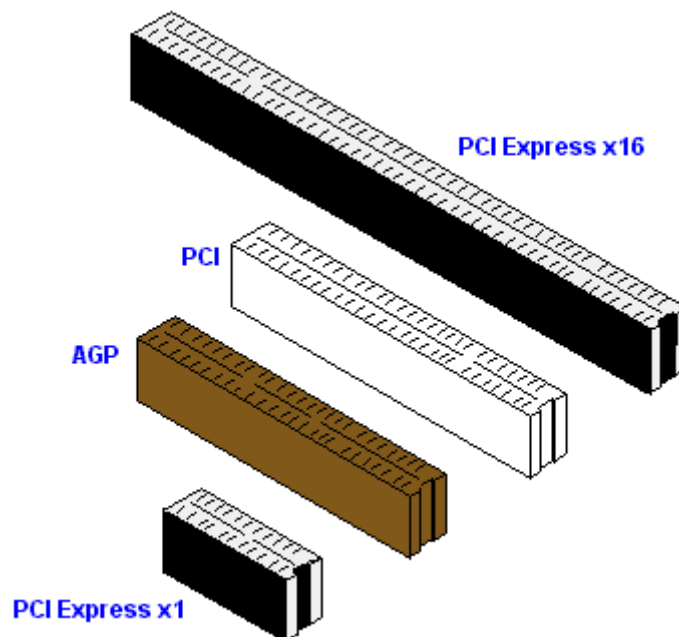


Fig.2.7:PCICardslottypes

The motherboard parts Accelerated Graphics Port (AGP) and (PCIe) focus on graphic cards only.

Random Access Memory (RAM):

It is the Computer main memory in which specific contents can be accessed. Two types of memory are possible with random-access circuits, static RAM (SRAM) and dynamic RAM (DRAM). A single memory chip is made up of several million memory cells.

ATA&SATA Hard Drives:

The Advanced Technology Attachment (ATA), (also known as IDE), is a standard interface for hard drives and optic drives (CD and DVD drives). The IDE has a low supported capacity. Hard drives with more than 500 MB could not be used with this interface.

The Serial ATA is the successor of the ATA interface; it is faster than ATA technology. It reduces cable bulk and cost. Faster data transfer through higher signalling rates, queuing protocol.

LEARNING OUTCOMES

Upon successful completion of the lab, students will be able to:

- LO1: Define & Differentiate between hardware and software.
- LO2: Define & Differentiate between Input and output devices and their functions.
- LO3: Define & Differentiate between application and system software.
- LO4: Students able to assemble and disassemble computer system components.

CONCLUSION:

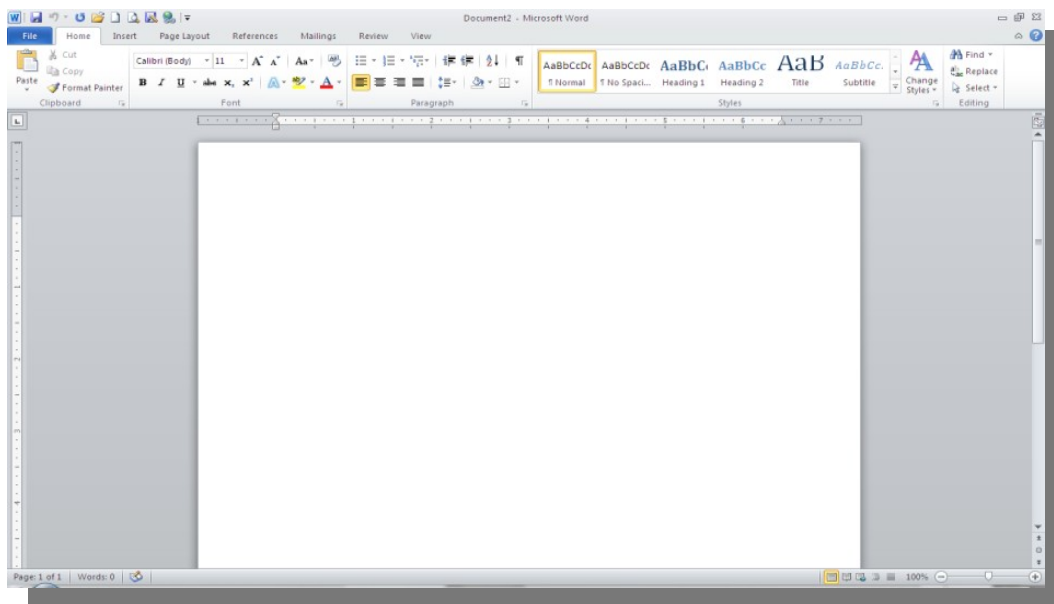
Lab#03

Understanding Ribbon inside Microsoft WORD, Microsoft EXCEL, Microsoft PowerPoint

Objective:

Understanding Ribbon inside Microsoft WORD, Microsoft EXCEL, Microsoft PowerPoint

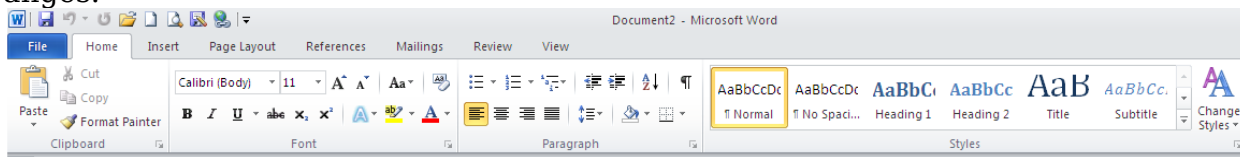
MSWORD



The Ribbon

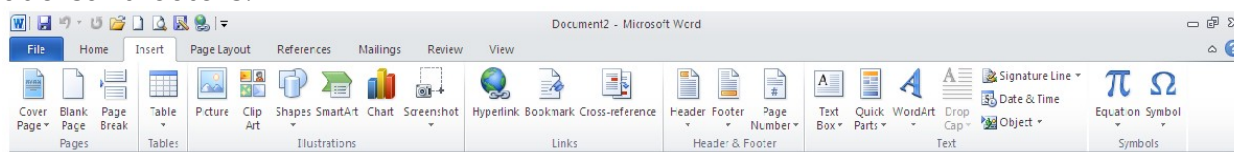
- **HomeTab:**

This is the most used tab; it incorporates all text formatting features such as font and paragraph changes.



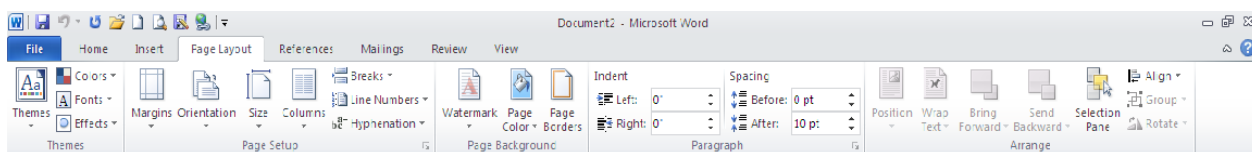
- **InsertTab:**

This tab allows you to insert a variety of items into a document from pictures, clipart, tables and headers and footers.



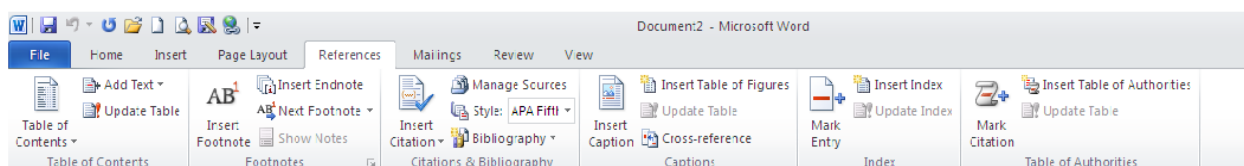
- **PageLayoutTab:**

This tab has commands to adjust page elements such as margins, orientation, inserting columns, page backgrounds and themes.



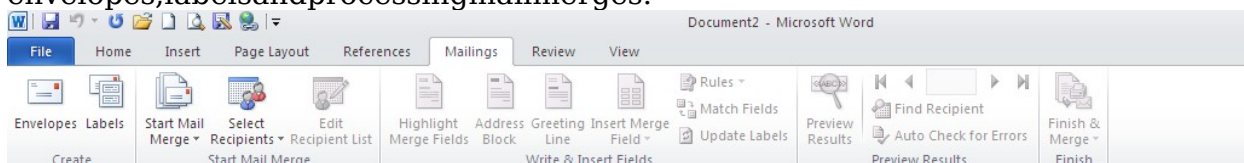
- **ReferenceTab:**

This tab has commands to use when creating a Table of Contents and citation page for a paper. It provides you with many simple solutions to create these typically difficult to produce documents.



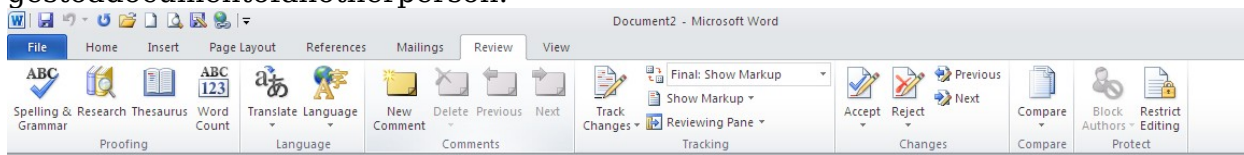
- **MailingTab:**

This tab allows you to create documents to help when sending out mailings such as printing envelopes, labels and processing mail merges.



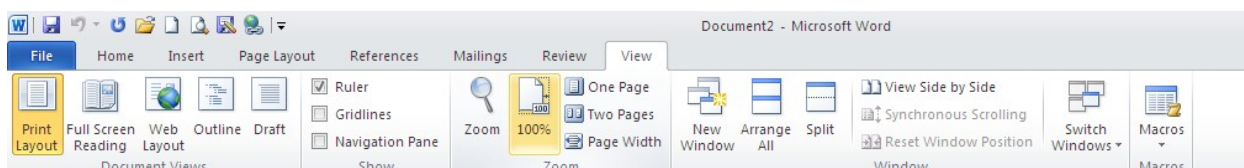
- **ReviewTab:**

This tab allows you to make any changes to your document due to spelling and grammar issues. It also holds the track changes feature which provides people with the ability to make notes and changes to a document of another person.

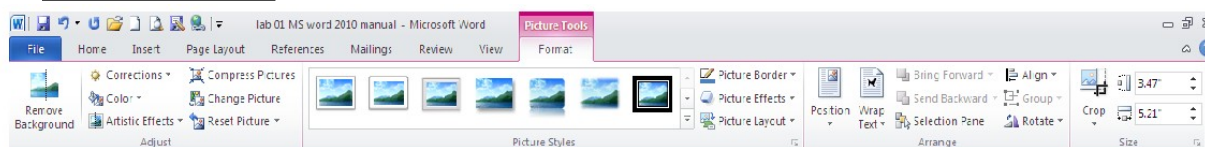


- **ViewTab:**

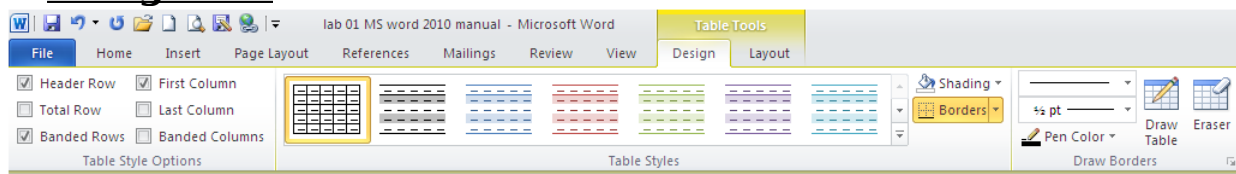
This tab allows you to change the view of your document to a different two-page document or zoom.



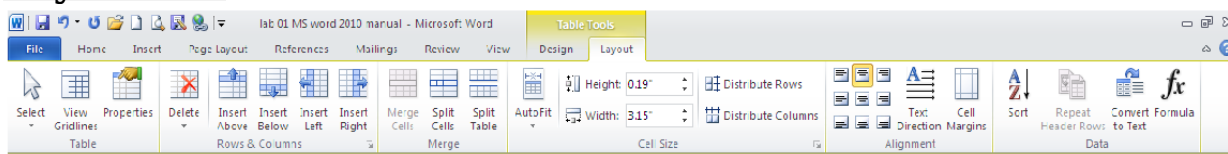
- **Formattab:**



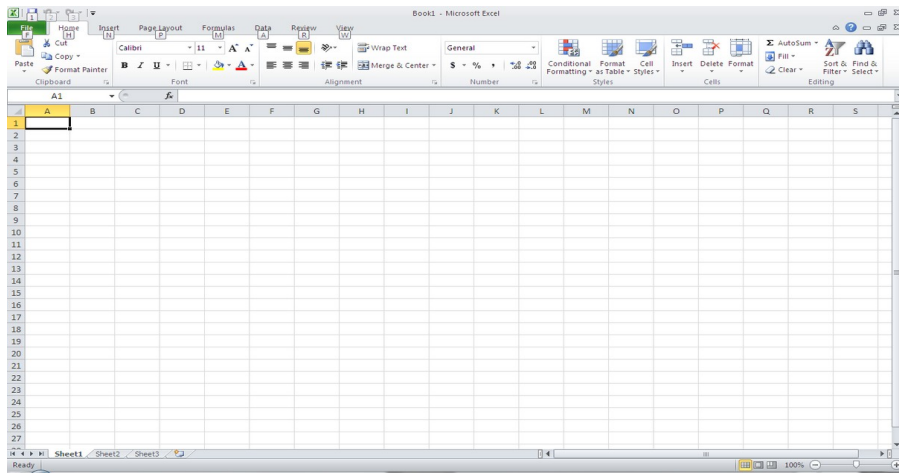
- **Design tab:**



- **Layouttab:**



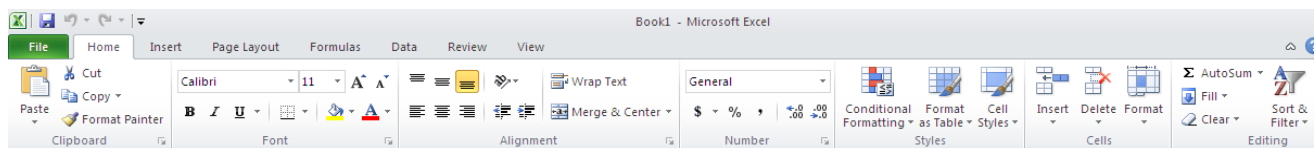
MSEXCEL



The Ribbon

- **HomeTab:**

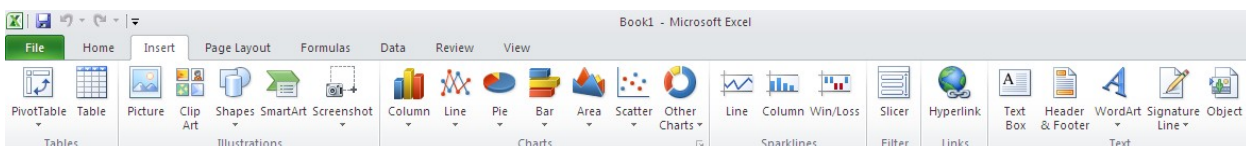
This is the most used tab; it incorporates all text and cell formatting features such as font and paragraph changes. The Home Tab also includes basic spreadsheet formatting elements such as text



wrap, merging cells and cell style.

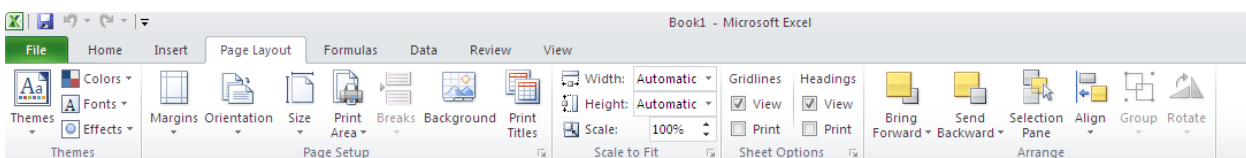
- **InsertTab:**

This tab allows you to insert a variety of items into a document from pictures, clipart, and headers and footers.



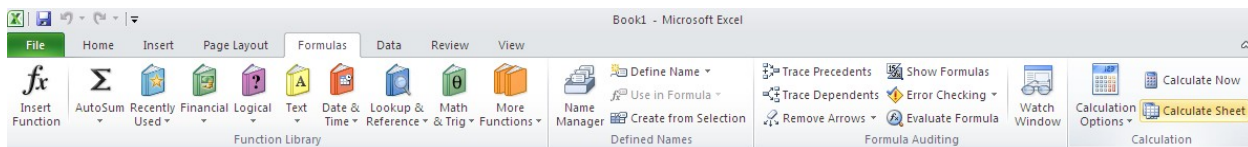
- **PageLayoutTab:**

This tab has commands to adjust pages such as margins, orientation, and themes.



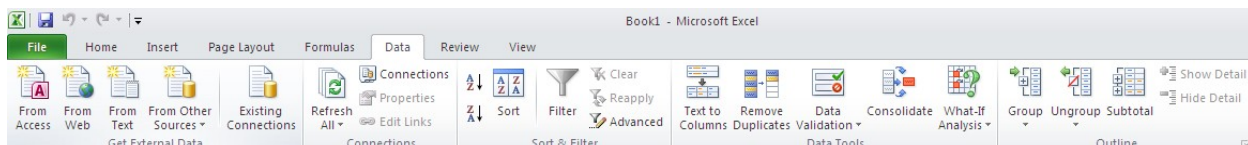
- **FormulasTab**

This tab has commands to use when creating Formulas. This tab holds an immense function library which can assist when creating any formula or function in your spreadsheet.



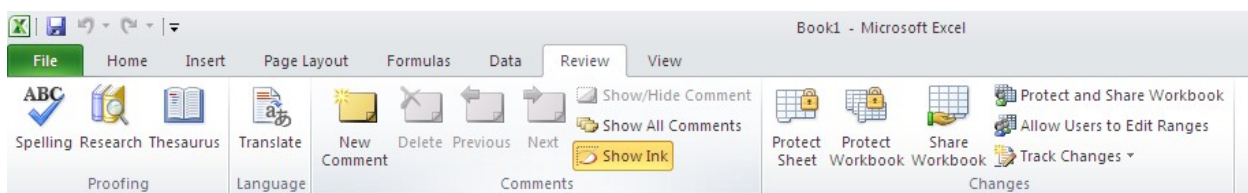
• **DataTab**

This tab allows you to modify worksheets with large amounts of data by sorting and filtering as well as analyzing and grouping data.



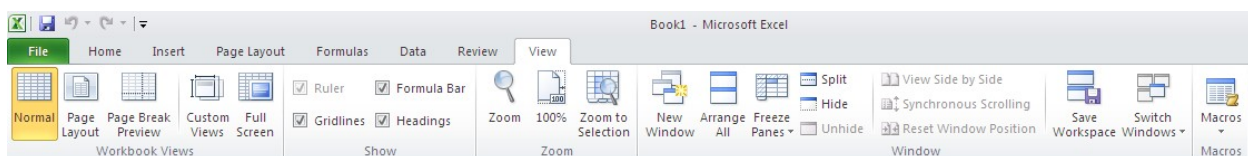
• **ReviewTab**

This tab allows you to correct spelling and grammar issues as well as set up security protections. It also provides the track changes and notes feature providing the ability to make notes and changes someone's document.



• **ViewTab**

This tab allows you to change the view of your document including freezing or splitting panes, view gridlines and hide cells.



MS PowerPoint

The Ribbon:

Tabs:

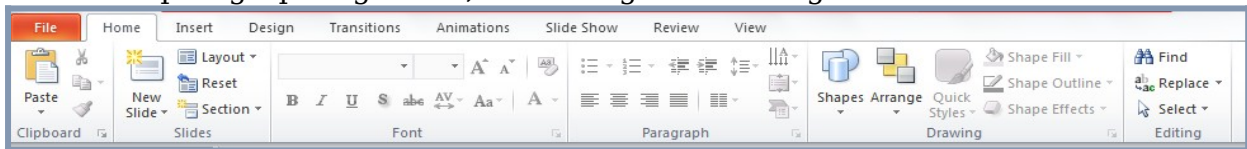
The Tabs represent different categories of functions on the ribbon.



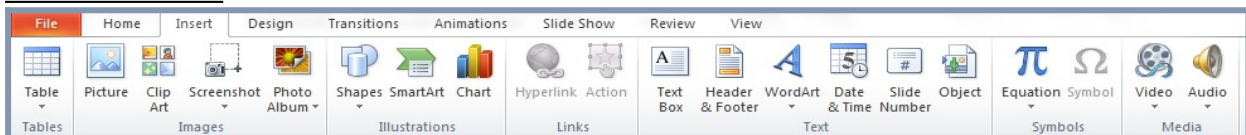
When you first start Microsoft PowerPoint, you normally see the Home tab.

- **Home Tab:**

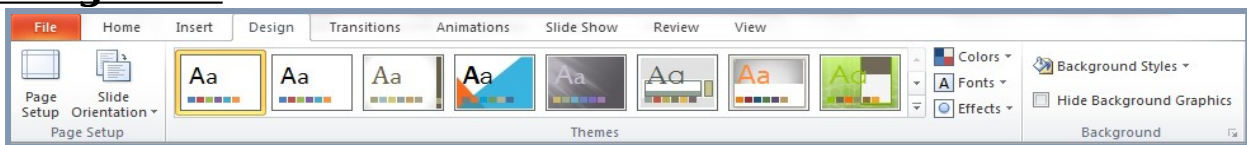
This tab contains icons and other controls that are most commonly used within Microsoft PowerPoint, such as making text display as bold or underlined. You can also find more advanced options such as paragraph alignment, numbering and bulleting.



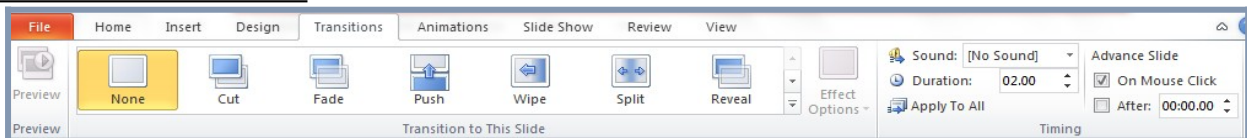
- **Insert Tab:**



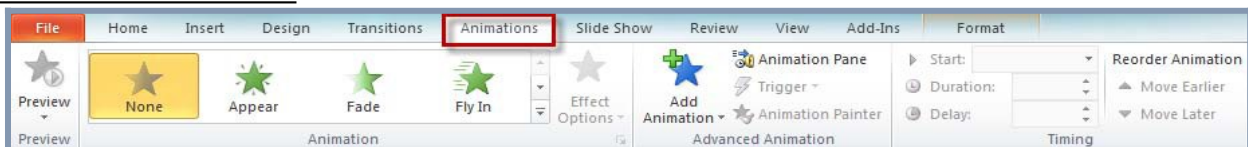
- **Design Tab:**



- **Transition Tab:**



- **Animation Tab:**



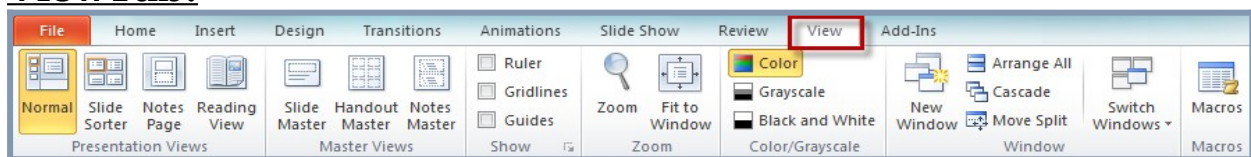
- **Slideshow Tab:**



- **ReviewTab:**



- **ViewTab:**

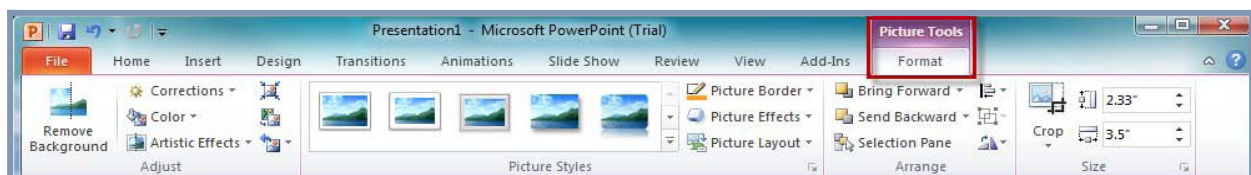


ContextualTabs

Whencertainobjectsareselectedotherondemandtabsoftenappear.

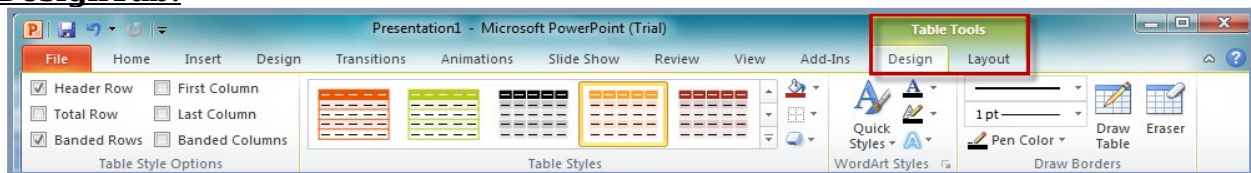
- **FormatTab:**

Inthisexamplewhenapictureisselectedonaslide,thePictureToolsFormattabappears.



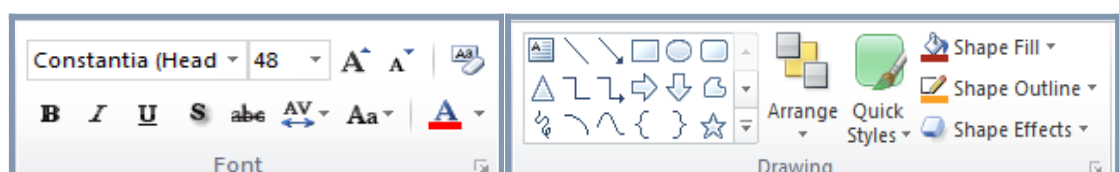
Inthisexamplewhenatableisselectedonaslidetherearetwocontextualtabs,DesignandLayout.

- **DesignTab:**



Groups:

Therearegroupsof toolson each ribbon. Commandsaregroupedbycategoryoneach ribbon.



Lab

**UnderstandingQuickAccess,StatusBar,DocumentsViews inMicro
softWORD**

Objective:

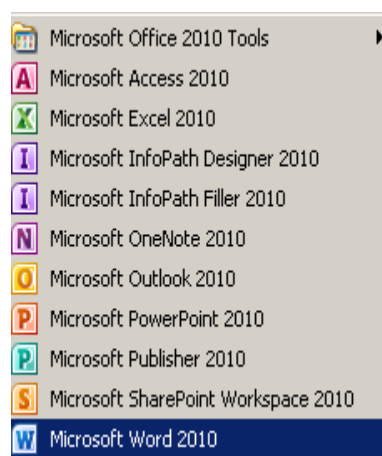
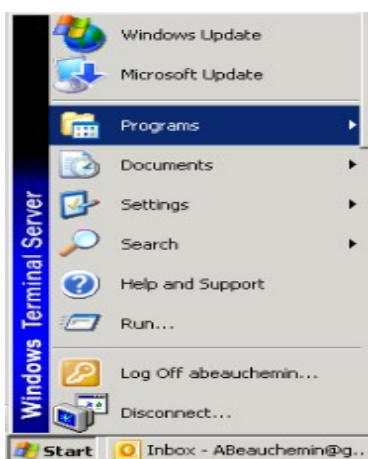
UnderstandingQuickAccess,StatusBar,DocumentsViews inMicrosoftWORD.

MSWord:

Microsoft Word is a word-processing program, designed to help you create professional-quality documents. With the finest document-formatting tools, Word helps you organize and write your documents more efficiently. Word also includes powerful editing and revising tools so that you can collaborate with others easily. It can be used to create documents such as letters, memos and faxes. The area where the text is written is known as Editor.

Getting Started

Now that you have an understanding of where things are located, let's look at the steps needed to create a document.



• Opening Outlook

You may have a shortcut to Word on your desktop, if so double click the icon and Word will open. If not

Follow the steps below:

1. Click on the Start button
2. Highlight Programs
3. Highlight Microsoft Office
4. Click on Microsoft Word 2010

The Quick Access Toolbar

The Quick Access toolbar provides you with access to commands you frequently use. By default Save, Undo, and Redo appear on the Quick Access toolbar.

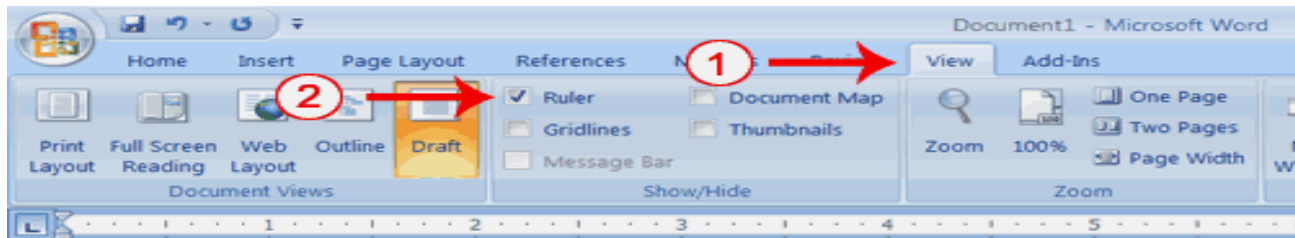
The Title Bar:

The Title bar displays the title of the document on which you are currently working.

The Ruler:

The ruler is found below the Ribbon. You can use the ruler to change the format of your document quickly. If your ruler is not visible, follow the steps listed here:

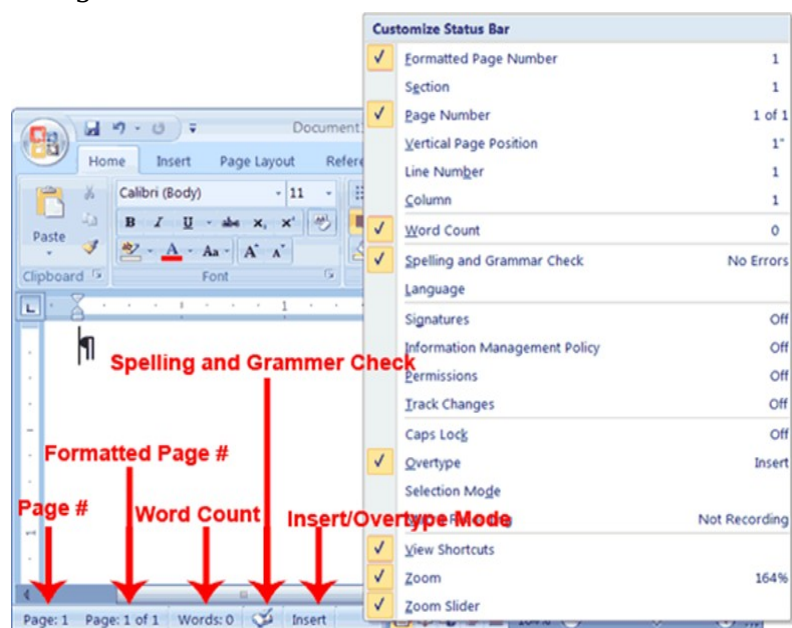
1. Click the View tab to choose it.
2. Click the checkbox next to Ruler in the Show/Hide group. The ruler appears below the Ribbon.



The Status Bar

The Status bar appears at the very bottom of your window and provides such information as the current page and the number of words in your document. You can change what displays on the Status bar by right-

clicking on the Status bar and selecting the options you want from the Customize Status Bar menu. You click a menu item to select it. You click it again to deselect it. A checkmark next to an item means it is selected.



Understanding Document Views

In Word 2010, you can display your document in one of five views: Draft, Web Layout, Print Layout, Full Screen Reading, or Online Layout.

- **Draft View**

Draft view is the most frequently used view. You use Draft view to quickly edit your document.

- **Web Layout**

Web Layout view enables you to see your document as it would appear in a browser such as Internet Explorer.

- **PrintLayout**

ThePrintLayoutviewshowsthedocumentasitwilllookwhenitisprinted.

- **ReadingLayout**

ReadingLayoutviewformatsyourscreentomakereadingyourdocumentmorecomfortable.

- **OutlineView**

Outlineviewdisplaysthedocumentinoutlineform.Youcandisplayheadingswithoutthe text.Ifyoumoveaheading,theaccompanyingtextmoveswithit.

SomeShortKeys:

s.no	Shortcutkey	press
1	Help	F1
2	Find&Replace	F5
3	Spelling&grammar	F7
4	Saveas	F12
5	Copy	Ctrl+c
6	Save	Ctrl+s
7	Open	Ctrl+o
8	Undo	Ctrl+d
9	Redo	Ctrl+y
10	New	Ctrl+n
11	Beginningofline	Home
12	Endofline	End
13	Bold	Ctrl+b
14	Italic	Ctrl+i
15	Underline	Ctrl+u
16	Growfontsize	Ctrl+>
17	Shrinkfont	Ctrl+<
18	Justify	Ctrl+j
19	Alignextright	Ctrl+r
20	Aligntextleft	Ctrl+l
21	Center	Ctrl+e
22	Find	Ctrl+f

Lab#

Creating, opening and formatting documents, basic operations on files and using different templates in Microsoft WORD

Objective:

Creating, opening and formatting documents, basic operations on files and using different templates in Microsoft WORD

Create a New Document:

Click the **File** tab and then click **New**.

Using Templates

Word 2010 allows you to apply built-in templates from a wide selection of popular Word templates, including resumes, agendas, business cards, and faxes.

To find and apply a template in Word, do the following:

1. On the **File** tab, click **New**.
2. Under **Available Templates**, do one of the following:
 - To use one of the built-in templates, click **Sample Templates**, click the template that you want, and then click **Create**.
 - To reuse a template that you've recently used, click **Recent Templates**, click the template that you want, and then click **Create**.
 - To find a template on Office.com, under **Office.com Templates**, click the template category that you want, click the template that you want, and click **download** to download the template from Office.com to your computer.
3. Once you have selected your template you can modify it in any way to create the document you want.

Opening a document

1. Click the **File** tab, and then click **Open**.
2. In the left pane of the **Open** dialog box, click the drive or folder that contains the document.
3. In the right pane of the **Open** dialog box, open the folder that contains the document that you want.
4. Click the document and then click **Open**.

Save a File

1. Click the File button. A menu appears.
2. Click Save. The Save As dialog box appears if you are saving your document for the first time.
3. Specify the correct folder in the Save In box.
4. Name your document.
5. Click Save.

Exit a file

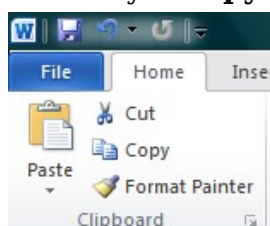
1. Click the file button, a menu appears.
2. Click on Exit option.

Cut, Copy and Paste:

If you would like to remove text from your document you can copy or cut the text from the document. Simply highlight the text and go to the **Home** tab in the **Clipboard** group and click **Cut** or **Copy**. You can also right click on your mouse and select **Cut** or **Copy**.

Pasting Text:

If you **Copy** text, you typically need to **Paste** it somewhere. The **Paste** feature in 2010 is much more detailed than in previous versions of Word. When you paste content, the **Paste Options** button provides different options, depending on the source of the content.



Keep Source Formatting: This option preserves the look of the original text.

Merge Formatting: This option changes the formatting so that it matches the text that surrounds it.

Keep Text Only: This option removes all the original formatting from the text.

Link & Keep Source Formatting: This option preserves the look of the original text, and it maintains a link to the source file and updates the pasted text with any changes that are made to the source file.

Link & Use Destination Styles: This option formats the text to match the style that's applied where the text is pasted. It also maintains a link to the source file and updates the pasted text with any changes that are made to the source file.

Picture: This option inserts the text as an image.

Use Destination Styles: This option formats the text to match the style that's applied where the text is pasted.

Use Destination Theme: This option formats the text to match the theme that's applied to the document where the text is pasted.

Undo:

If you make an error in your document click on the **Undo** command and it will remove the last thing you did.

Show/Hide Formatting Marks

The Show/

Hide command allows you to see every time you hit the spacebar, hit enter or tab. This feature can be quite useful when creating documents to understand where everything is placed within your document and see if any errors have been made.



Formatting Text

Formatting a document can range from modifying text size to adding graphics. It is easy to add creative touches to any document with the options Microsoft Word has to offer.

Modifying Fonts

The **Font** Group allows you to change your text font style, size, color and many other elements.

Change Text Case

You can change the case of selected text in a document by clicking a single button called **Change Case** on the ribbon.

- Choose an option from the dropdown list, which includes **Sentence case, lowercase, UPPERCASE, Capitalize Each Word, and TOGGLE CASE**.

Alternate way of changing text case

- Select the text.
- Now use the **Shift+F3** to change to the next case option.

Adding text effects

1. Select the text that you want to add an effect to.
2. On the **Home** tab, in the **Font** group, click **Text Effect**.
3. Click the effect that you want. For more choices, point to **Outline, Shadow, Reflection, or Glow**, and then click the effect that you want to add.

Remove text effects

Select the text that you want to remove an effect from. On the **Home** tab, in the **Font** group, click **Clear Formatting**.

Format Painter

The **Format Painter** feature allows you to quickly copy a format that you have applied to text already in your document.

1. Select the text or graphic that has the formatting that you want to copy.
2. On the **Home** tab, in the **Clipboard** group, single click **Format Painter**. The pointer will change to a paintbrush icon.
3. Bring your cursor to the text or graphic that you want to format and click on the text.
4. To stop formatting, press **ESC** or click on the **Format Painter** command again.

Clear Formatting

To get rid of all the styles, text effects, and font formatting in your document, do the following:

1. Select the text that you want to clear the formatting from. Or press **CTRL+A** to select everything in the document.
2. On the **Home** tab, in the **Font** group, click **Clear Formatting**.

Formatting Documents

- **Adjusting Line Spacing**
 - ❖ The default spacing is 1.15 lines spacing.
 - ❖ The easiest way to change the line spacing for an entire document is to highlight the paragraphs or entire document that you want to change the line spacing on.
 - ❖ On the **Home** tab, in the **Paragraph** group, click **Line Spacing**.

Lab

Understanding pagesetup and paragraph setting inside MSWORD documents

Objective:

Understanding pagesetup and paragraph setting inside MSWORD documents

Page Orientation:

You can choose either portrait (vertical) or landscape (horizontal) orientation for all or part of your document.

Change Page Orientation

1. On the Page Layout tab, in the Page Setup group, click Orientation.
2. Click Portrait or Landscape.

Page Margins

Page margins are the blank space around the edges of the page. In general, you insert text and graphics in the printable area inside the margins. When you change a document's page margins, you change where text and graphics appear on each page. You can change the page margin in either by choosing from one of Word's predefined settings.

Page Breaks

Word automatically inserts a page break when you reach the end of a page. If you want the page to break in a different place, you can insert a manual page break.

Inserting a Page Break

1. Click where you want to start a new page.
2. On the Insert tab, in the Pages group, click Page Break.

Deleting a Page Break

You cannot delete the page breaks that Word inserts automatically; you can only delete a page break that you insert manually.

1. Go to the page break you would like to remove.
2. Select the page break by clicking in the margin next to the dotted line.
3. Press the DELETE key on your keyboard.

Headers, Footers & Page Number:

You can add headers, footers and page numbers numerous ways. The simplest way is to double-click on the top or bottom of the page and the header and footer area will appear. Enter the text you wish to be displayed at the top or bottom of every page.

Add Page Numbers:

If you want a page number on each page, you can quickly add a page number from the gallery.

1. On the Insert tab, in the Header & Footer group, click Page Number.
2. Click the page number location that you want.
3. In the gallery, scroll through the options, and then click the page number format that you want.
4. To return to the body of your document, click Close Header and Footer on the Design tab (under Header & Footer Tools).

Add Header or Footer

1. On the Insert tab, in the Header & Footer group, click Header or Footer.
2. Click the header or footer that you want to add to your document and your header or footer area will open.
3. Type text in the header or footer area.
4. To return to the body of your document, click Close Header and Footer on the Design tab (under Header & Footer Tools).

Remove page numbers, headers, and footers

1. Click on the Header, Footer or Page Number Command.
2. A dropdown menu will appear.
3. Click Remove at the bottom of the menu.

Bulleted or Numbered List

You can quickly add bullets or numbers to existing lines of text, or Word can automatically create lists as you type. By default, if you start a paragraph with an asterisk or a number 1., Word recognizes that you are trying to start a bulleted or numbered list.

Insert Bulleted or Numbered List

1. Click on the area where you would like your list to appear or highlight the text you would like to be in a list.
2. Go to the Home tab, in the Paragraph group, click Bullets or Numbering.
3. A bullet(s) or number(s) will be inserted.

Select Bullets or Numbering Style

1. Select the item that you want to add bullets or numbering to.
2. On the Home tab, in the Paragraph group, click the arrow next to the Bullets or Numbering command.
3. Select the bullet or number format you would like to be inserted.

Lab

Learning graphics and understanding use of SmartArt, Charts, Tables and Word Art in MSWORD

Objective:

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Working with Graphics

- **Inserting Shapes**

You can add one shape to your file or combine multiple shapes to make a drawing or a more complex shape. Available shapes include lines, basic geometric shapes, arrows, equation shapes, flowchart shapes, stars, banners, and callouts. After you add one or more shapes, you can add text, bullets, numbering, and QuickStyle to them.

1. On the **Insert** tab, in the **Illustrations** group, click **Shapes**.
2. A dropdown menu will appear, click the shape that you want.
3. Click anywhere in the document, and then drag to place the shape.

- **Insert Text to Shapes**

Once you have added a shape, you may want to add text inside the shape. All you have to do is click on the inside of the shape and start typing.

- **Format Shapes**

After you insert a shape a new tab called **Drawing Tools Format** will appear every time you click on the shape.

1. Click the shape that you want to apply a new or different QuickStyle to.
2. Go to the **Drawing Tools Format** tab, in the **Shape Styles** group, click the style that you want to be applied.
3. To see more QuickStyles, click the **More** button. The **Drawing Tools Format** Tab also allows you to change the shape fill, outline, and effects and select how the text in your document is wrapped around the shape.

- **Delete Shapes**

If you decide you no longer want the shape in your document then click on the shape and then press DELETE.

- **Inserting Text Boxes**

A text box is an object that lets you put and type text anywhere in your file.

1. On the **Insert** tab, in the **Text** group, click **Text Box** and a dropdown menu will appear.
2. Click on a text box template or click **Draw Text Box** at the bottom of the dropdown menu to draw your own text box.
3. If you elect to draw your own text box you need to click in the document, and then drag to draw the text box the size that you want.

4. To add text to a text box, click inside the text box, and then type or paste text.

- To format text in the text box, select the text, and then use the formatting options in the **Font** group on the **Home** tab.
- To position the text box, click it, and then when the pointer becomes a , drag the text box to a new location.

- **Deleting Text Boxes**

To remove a text box, just click the border of the text box that you want to delete, and then press DELETE. Make sure that the pointer is not inside the text box, but rather on the border of the text box. If the pointer is not on the border, pressing DELETE will delete the text inside the text box and not the text box.

WordArt

WordArt can be used to add special text effects to your document. For example, you can stretch a title, skew text, make text fit a preset shape, or apply a gradient fill. This WordArt becomes an object that you can move or position in your document to add decoration or emphasis. You can modify or add to the text in an existing WordArt object whenever you want. To add WordArt to text in your document, complete the following steps:

1. On the **Insert** tab, in the **Text** group, click **WordArt**.
2. A Drop-down menu will appear, click the WordArt style that you want.
3. A Text Box will appear with the words "Enter your text here", Enter your text.

- **Insert Picture/ClipArt**

Pictures and clip art can be inserted or copied into a document from many different sources, including downloaded from a clip art Website provider, copied from a Web page, or inserted from a folder where you save pictures.

- **Insert ClipArt**

1. On the **Insert** tab, in the **Illustrations** group, click **ClipArt**.
2. A **ClipArt** task pane will appear on the right of your screen, in the **Search for** box, type a word or phrase that describes the clip art that you want.
3. Click **Go**.
4. In the list of results, double-click on the clip art to insert it into your document.

- **Insert Picture from Web**

1. Open the document.
2. From the Web page, drag the picture that you want into the Word document.

- **InsertPicturefromFile**

To insert a picture saved in your computer, insert it by following these steps.

1. Click where you want to insert the picture in your document.
2. On the **Insert** tab, in the **Illustrations** group, click **Picture**.
3. Locate the picture that you want to insert. For example, you might have a picture file located in **My Documents**.
4. Double-click the picture that you want to insert and it will appear in your document.

- **CroppingaPicture**

Cropping reduces the size of a picture by removing vertical or horizontal edges. Cropping is often used to hide or trim a part of a picture, either for emphasis or to remove unwanted portions.

1. Click on the picture that you want to crop.
2. Go to **Picture Tools**, on the **Format** tab, in the **Size** group, click **Crop**.
3. Black dotted lines will appear around your picture. Then drag the center cropping handle on the side inward.
4. As you drag the cropping handle you will notice the area of your graphic you want removed will become gray.
5. Once you have cropped out everything you want, click outside of the graphic for the gray area you want removed to disappear.

- **Uncrop a Picture**

You can always restore a resized or a cropped picture to its original appearance.

1. Click on your picture
2. Go to **Picture Tools**, on the **Format** tab, in the **Size** group, click **Crop**.
3. Black dotted lines will appear around your picture. Drag the black lines away from the center of the picture and the original picture will appear.

Advanced Formatting Techniques

- **Create Columns**

Columns can be used in documents such as brochures, newsletters or to save space when creating lists. Add columns before entering text:

1. Go to the **Page Layout** tab, in the **Page Setup** group, click **Columns**.
2. Click the layout that you want. Your document will be formatted in columns

- **Add columnsto part of a document**

To do that highlight the text you want formatted in columns, or place your cursor where you want columns to begin.

1. On the **Page Layout** tab, in the **Page Setup** group, click **Columns**.
2. Click **More Columns**.
3. Click the number of columns that you want.
4. In the **Apply to** list, click **Selected text** or **This point forward**.

SmartArt Graphic

A SmartArt graphic is a visual representation of your information that you can quickly and easily create, choosing from among many different layouts, to effectively communicate your message or ideas. You can create SmartArt graphics in Excel, Outlook, PowerPoint, and Word.

SmartArt graphics enable you to create designer-quality illustrations with only a few clicks of your mouse. When you create a SmartArt graphic, you are prompted to choose a type of SmartArt graphic, such as **Process**, **Hierarchy**, **Cycle**, or **Relationship**. Each type of SmartArt graphic contains several different layouts. After you choose a layout, it is easy to switch the layout or type of a SmartArt graphic. Most of your text and other content, colors, styles, effects, and text formatting are automatically carried over to the new layout.

When you select a layout, placeholder text (such as **[Text]**) is displayed, so that you can see how your SmartArt graphic looks, nor is it displayed during a slideshow. However, the shapes are always displayed and printed, unless you delete them. You can replace the placeholder text with your own content.

- **Create a SmartArt Graphic**

1. On the **Insert** tab, in the **Illustrations** group, click **SmartArt**.
2. In the **Choose a SmartArt Graphic** dialog box, click the type and layout that you want.
3. Enter your text by doing one of the following:
 - Click **[Text]** in the Text pane, and then type your text.
 - Copy text from another location, click **[Text]** in the Text pane, and then paste your text.

- **Add or Delete Shapes in SmartArt Graphic**

1. Click the SmartArt graphic that you want to add another shape to.
2. Click the existing shape that is located closest to where you want to add the new shape.
3. Under **SmartArt Tools**, on the **Design** tab, in the **Create Graphic** group, click the arrow under **Add Shape**.
4. Do one of the following:
 - To insert a shape after the selected shape, click **Add Shape After**.
 - To insert a shape before the selected shape, click **Add Shape Before**.

- **Format SmartArt Graphic**

You can apply color variations to the shapes in your SmartArt graphic.

1. Click your SmartArt graphic.

2. Under **SmartArtTools**, on the **Design** tab, in the **SmartArtStyles** group,
3. Select the SmartArt Style you wish to apply to add line styles, bevels or 3-D effects.
4. In the **SmartArtStyles** group you can also click **Change Colors** to further modify your SmartArt graphic.

Tables

Using tables in Word can provide you with additional elements to any document. Tables can be used to create lists or format text in an organized fashion.

- **Inserting a Table**

1. Click where you want to insert a table.
2. On the **Insert** tab, in the **Tables** group, click **Table**
3. A drop-down box will appear; click and hold your mouse then drag to select the number of rows and columns that you want inserted into your document. You will see your table appearing in your document as you drag on the grid.
4. Once you have highlighted the rows and columns you would like to go, move your mouse and the table will be in your document.

- **Add Row/Column to Table**

1. Click on the table.
2. Under **TableTools**, go to the **Layout** tab
3. Click on the **Insert Above** or **Insert Below** to add a row, Click on **Insert Left** or **Insert Right** to insert a column.
4. Click on **Delete** to remove a column, row or cell.

- **Delete a Table**

1. Rest the pointer on the table until the table move handle appears, and then click the table move handle.
2. Press BACKSPACE on your keyboard.

Finalizing a Document

- **Using the "Spell Check" Feature**

As you type your document, red wavy lines will appear under any word that is spelled incorrectly. The fastest way to fix spelling errors is to:

1. Put your cursor over the misspelled word and right-click.
2. A drop-down box will appear with correct spellings of the word.
3. Highlight and left-click the word you want to replace the incorrect word with.

To complete a more comprehensive Spelling and Grammar check, you can use the Spelling and Grammar feature.

1. Click on the **Review** tab
2. Click on the **Spelling & Grammar** command (a blue checkmark with ABC above it).
3. A **Spelling and Grammar** box will appear.
4. You can correct any Spelling or Grammar issue within the box.

Lab

Understanding spreadsheets and workbook in Microsoft EXCEL

Objective:

Understanding spreadsheets and workbook in Microsoft EXCEL.

MICROSOFT EXCEL:

Excel is a spreadsheet program in the Microsoft Office system. You can use Excel to create and format workbooks (a collection of spreadsheets) in order to analyze data and make more informed business decisions. Specifically, you can use Excel to track data, build models for analyzing data, write formulas to perform calculations on that data, pivot the data in numerous ways, and present data in a variety of professional-looking charts.

Getting Started

Now that you have an understanding of where things are located, let's look at the steps needed to create an Excel document.

Opening Outlook

You may have a shortcut on your desktop, if so double click the icon and Excel will open. If not follow the steps below:

1. Click on the Start button
2. Highlight Programs
3. Highlight Microsoft Office
4. Click on Microsoft Excel 2010

Create a New Workbook

1. Click the **File** tab and then click **New**.
2. Under **Available Templates**, double click **Blank Workbook** or Click **Create**.

Enter Data in a Worksheet

1. Click the cell where you want to enter data.
2. Type the data in the cell.
3. Press Enter or Tab to move to the next cell.

Modifying Spreadsheets

In order to create an understandable and professional document you will need to make adjustments to the cells, rows, columns and text. Use the following processes to assist when creating a spreadsheet.

Cut, Copy, and Paste Data

You can use the Cut, Copy, and Paste commands in Microsoft Office Excel to move or copy entire cells or their contents.

NOTE: Excel displays an animated moving border around cells that have been cut or copied. To cancel a moving border, press ESC.

Move/Copy Cells

When you move or copy a cell, Excel moves or copies the entire cell, including formulas and their resulting values, cell formats, and comments.

1. Select the cell(s) that you want to move or copy.
2. On the **Home** tab, in the **Clipboard** group, do one of the following:
 - a. To move cells, click **Cut**.
 - b. To copy cells, click **Copy**.
3. Click in the center of the cell you would like to paste the information to.
4. On the **Home** tab, in the **Clipboard** group, click **Paste**.

Column Width and Row Height

On a worksheet, you can specify a column width of 0 to 255 and a row height of 0 to 409. This value represents the number of characters that can be displayed in a cell that is formatted with the standard font. The default column width is 8.43 characters and the default row height is 12.75 points. If a column/row has a width of 0, it is hidden.

Set Column/Row Width/Height

1. Select the column(s) or row(s) that you want to change.
2. On the **Home** tab, in the **Cells** group, click **Format**.
3. Under **Cell Size**, click **Column Width** or **Row Height**.
4. A **Column Width** or **Row Height** box will appear.
5. In the **Column Width** or **Row Height** box, type the value that you want your column or row to be.

Merge or Split Cells

When you merge two or more adjacent horizontal or vertical cells, the cells become one larger cell that is displayed across multiple columns or rows. When you merge multiple cells, the contents of only one cell appear in the merged cell.

Merge and Center Cells

1. Select two or more adjacent cells that you want to merge.
2. On the Home tab, in the Alignment group, click Merge and Center.
3. The cells will be merged in a row or column, and the cell contents will be centered in the merged cell.

Merge Cells

To merge cells only, click the arrow next to Merge and Center, and then click Merge Across or Merge Cells.

Split Cells

1. Select the merged cell you want to split.
2. To split the merged cell, click Merge and Center. The cell will split and the contents of the merged cell will appear in the upper-left cell of the range of split cells.

Automatically Fill Data

To quickly fill in several types of data series, you can select cells and drag the fill handle. To use the fill handle, you select the cell that you want to use as a basis for filling additional cells, and then drag the fill handle across or down the cells that you want to fill.

1. Select the cell that contains the formula that you want to be brought to other cells.
2. Move your cursor to the small black square in the lower-right corner of a selected cell, also known as the fill handle. Your pointer will change to a small black cross.
3. Click and hold your mouse, then drag the fill handle across the cells, horizontally to the right or vertically down, that you want to fill.
4. The cells you want filled will have a gray-looking border around them. Once you fill all of the cells, let go of your mouse and your cells will be populated.

Lab

FormattingspreadsheetsandunderstandingBasicformulasinMicrosoftEXCEL

Objectives:

FormattingspreadsheetsandunderstandingBasicformulasinMicrosoftEXCEL

FormattingSpreadsheets

To further enhance your spreadsheet, you can format a number of elements such as text, numbers, coloring, and table styles. Spreadsheets can become professional documents used for company meetings or can even be published.

WrapText

You can display multiple lines of text inside a cell by wrapping the text. Wrapping text in a cell does not affect other cells.

1. Click the cell in which you want to wrap the text.
2. On the **Home** tab, in the **Alignment** group, click **WrapText**.
3. The text in your cell will be wrapped.

NOTE: If the text is a long word, the characters won't wrap (the word won't be split); instead, you can widen the column or decrease the font size to see all the text. If all the text is not visible after you wrap the text, you might have to adjust the height of the row. On the **Home** tab, in the **Cells** group, click **Format**, and then under **Cell Size** click **AutoFit Row**.

Format Numbers

In Excel, the format of a cell is separate from the data that is stored in the cell. This display difference can have a significant effect when the data is numeric. For example, numbers in cells will default as rounded numbers, date and time may not appear as anticipated. After you type numbers in a cell, you can change the format in which they are displayed to ensure the numbers in your spreadsheet are displayed as you intended.

1. Click the cell(s) that contain the numbers that you want to format.
2. On the **Home** tab, in the **Number** group, click the arrow next to the **Number Format** box, and then click the format that you want.

If you are unable to format numbers in the detail you would like that you can click on the **More Number Formats** at the bottom of the **Number Format** dropdown list.

In the Category list, click the format that you want to use, and then adjust settings to the right of the Format Cells dialog box. **For example**, if you're using the Currency format, you can select a different currency symbol, show more or fewer decimal places, or change the way negative numbers are displayed.

Cell Borders

By using predefined border styles, you can quickly add a border around cells or ranges of cells. If predefined cell borders do not meet your needs, you can create a custom border.

Apply Cell Borders

1. On a worksheet, select the cell or range of cells that you want to add a border to, change the border style on, or remove a border from.
2. Go to the **Home** tab, in the **Font** group
3. Click the arrow next to **Borders**
4. Click on the border style you would like
5. The border will be applied to the cell or cell range

Remove Cell Borders

1. Go to the **Home** tab, in the **Font** group
2. Click the arrow next to **Borders**
3. Click **No Border**.

Cell Styles

You can create a cell style that includes a custom border, colors and accounting formatting.

1. On the **Home** tab, in the **Styles** group, click **Cell Styles**.
2. Select the different cell style option you would like applied to your spreadsheet.

Cell and Text Coloring

You can also modify a variety of cell and text colors manually.

Cell Fill

1. Select the cell that you want to apply or remove a fill color from.
2. Go to the **Home** tab, in the **Font** group and select one of the following options:
 - a. To fill cells with a solid color, click the arrow next to **Fill Color**, and then under **Theme Colors** or **Standard Colors**, click the color that you want.
 - b. To fill cells with a custom color, click the arrow next to **Fill Color**, click **More Colors**, and then in the **Colors** dialog box select the color that you want.
 - c. To apply the most recently selected color, click **Fill Color**.

Remove Cell Fill

1. Select the cell that contains a fill color or fill pattern.
2. On the **Home** tab, in the **Font** group, click the arrow next to **Fill Color**, and then click **No Fill**.

Text Color

1. Select the cell, range of cells, text, or character that you want to format with a different text color.
2. On the **Home** tab, in the **Font** group and select one of the following options:
 - a. To apply the most recently selected text color, click **Font Color**.
 - b. To change the text color, click the arrow next to **Font Color**, and then under **Theme Colors** or **Standard Colors**, click the color that you want to use.

Bold, Underline and Italics Text

1. Select the cell, range of cells, or text.
2. Go to the **Home** tab, in the **Font** group

3. Click on the Bold (**B**) Italic (*I*) or Underline (U) commands.
4. The selected command will be applied.

Customize Worksheet Tab

1. On the **Sheet tab** bar, right-click the sheet tab that you want to customize
2. Click **Rename** to rename the sheet or **Tab Color** to select a tab color.
3. Type in the name or select a color you would like for your spreadsheet.
4. The information will be added to the tab at the bottom of the spreadsheet.

Formulas in Excel

Formulas are equations that perform calculations on values in your worksheet. A formula always starts with an equals sign (=). An example of a simple is **=5+2*3** that multiplies two numbers and then adds a number to the result. Microsoft Office Excel follows the standard order of mathematical operations. In the preceding example, the multiplication operation (2*3) is performed first, and then 5 is added to its result.

You can also create a formula by using a function which is a prewritten formula that takes a value, performs an operation and returns a value. **For example, the formulas =SUM(A1:A2) and SUM(A1,A2) both use the SUM function to add the values in cells A1 and A2.**

Depending on the type of formula that you create, a formula can contain any or all of the following parts.

The diagram shows the formula **=PI()*A2^2** with four numbered callouts in colored circles: 1 (blue) points to the equals sign, 2 (green) points to the PI function, 3 (red) points to the multiplication sign, and 4 (blue) points to the exponentiation sign. A red bracket is placed under the **A2^2** portion of the formula.

Functions A function, such as **PI()** or **SUM()**, starts with an equals sign (=).

Cell references

You can refer to data in worksheet cells by including cell references in the formula. **For example,** the cell reference **A2** returns the value of that cell or uses that value in the calculation.

Constants

You can also enter constants, such as numbers (such as **2**) or text values, directly into a formula.

Operators

Operators are the symbols that are used to specify the type of calculation that you want the formula to perform.

Create a Simple Formula

1. Click the cell in which you want to enter the formula.
2. Type **=** (equals sign).
3. Enter the formula by typing the constants and operators that you want to use in the calculation.
4. Press **ENTER**.

EXAMPLE FORMULA	WHAT IT DOES
=5+2	Adds 5 and 2
=5-2	Subtracts 2 from 5
=5/2	Divides 5 by 2
=5*2	Multiplies 5 times 2
=5^2	Raises 5 to the 2nd power

Create a Formula with Cell References

- The first cell reference is B3, the color is blue, and the cell range has a blue border with square corners.
- The second cell reference is C3, the color is green, and the cell range has a green border with square corners.

To create your formula:

1. Click the cell in which you want to enter the formula.
2. In the **formulabar**, at the top of the Excel window that you use, type **=** (equals sign).
3. Click on the 1st cell you want in the formula.
4. Enter an Operator such as +, or *.
5. Click on the next cell you want in the formula. Continue steps 3–5 until the formula is complete.
6. Hit the **ENTER** key on your keyboard.

EXAMPLE FORMULA	WHAT IT DOES
=A1+A2	Adds the values in cells A1 and A2
=A1-A2	Subtracts the value in cell A2 from the value in A1
=A1/A2	Divides the value in cell A1 by the value in A2
=A1*A2	Multiplies the value in cell A1 times the value in A2
=A1^A2	Raises the value in cell A1 to the exponential value specified in A2

Create a Formula with Function

1. Click the cell in which you want to enter the formula.
 2. Click **Insert Function** on the formula bar. Excel inserts the equals sign(=) for you.
 3. Select the function that you want to use.
1. Enter the arguments.
 5. After you complete the formula, press ENTER

Use AutoSum

To summarize values quickly, you can also use **AutoSum**.

1. Select the cell where you would like your formula's solution to appear.
2. Go to the **Home** tab, in the **Editing** group,
3. Click **AutoSum**, to sum your numbers or click the arrow next to **AutoSum** to select a function that you want to apply.

Delete a Formula

When you delete a formula, the resulting values of the formula is also deleted. However, you can instead remove the formula only and leave the resulting value of the formula displayed in the cell.

- To delete formulas along with their resulting values, do the following:
 1. Select the cell or range of cells that contain the formula.
 2. Press DELETE.
- To delete formulas without removing their resulting values, do the following:
 1. Select the cell or range of cells that contain the formula.
 2. On the **Home** tab, in the **Clipboard** group, click **Copy**.
 3. On the **Home** tab, in the **Clipboard** group, click the arrow below **Paste**, and then click **Paste Values**.

Assignment

1. Make a marksheet on excel, and find more formulas also apply them.

Lab

Creating Charts and using advanced spreadsheet modification

Objective:

To become familiar with the **MSEXCEL** Environment.

- **How to make charts in excel.**
- **How to print workbook sheet.**

Charts in Excel

Microsoft Excel no longer provides the chart wizard. Instead, you can create a basic chart by clicking the chart type that you want on the Insert tab in the Charts group. Charts are used to display series of numeric data in a graphical format to make it easier to understand large quantities of data and the relationship between different series of data.

To create a chart in Excel, you start by entering the numeric data for the chart on a worksheet. Then you can plot that data into a chart by selecting the chart type that you want to use on the Insert tab, in the Charts group. Worksheet data Chart created from worksheet data

A chart has many elements. Some of these elements are displayed by default, others can be added as needed. You can change the display of the chart elements by moving them to other locations in the chart, resizing them, or by changing the format. You can also remove chart elements that you do not want to display.

1. The **chart area** is the entire chart and all its elements
2. The **plot area** is the area of the chart bounded by the axes.
3. The **data points** are individual values plotted in a chart represented by bars, columns, lines, or pies.
4. The **horizontal** (category) and **vertical** (value) **axis** along which the data is plotted in the chart.
5. The **legend** identifies the patterns or colors that are assigned to the data series or categories in the chart.
6. A **chart and axis title** are descriptive text that for the axis or chart.
7. A **data label** provides additional information about a data marker that you can use to identify the details of a data point in a data series.

Create a Chart

1. On the worksheet, arrange the data that you want to plot in a chart. The data can be arranged in rows or columns—Excel automatically determines the best way to plot the data in the chart.
2. Select the cells that contain the data that you want to use for the chart.

NOTE: If the cells that you want to plot in a chart are not in a continuous range, you can select nonadjacent cells or ranges as long as the selection forms a rectangle. You can also hide the rows or columns that you do not want to plot in the chart.

3. Go to the **Insert** tab, in the **Charts**

4. Click the chart type, and then click a chart subtype from the drop menu that will appear.

5. Click anywhere in the embedded chart to activate it. When you click on the chart, **Chart Tools** will be displayed which includes the **Design**, **Layout**, and **Format** tabs.

6. The chart will automatically be embedded in the worksheet. A chart name will automatically be assigned.

Move Chart to New Sheet

1. On the **Design** tab, in the **Location** group, click **Move Chart**.

2. Under **Choose where you want the chart to be placed**, click on the **Newsheet** bubble

3. Type a chart name in the **Newsheet** box.

Change Chart Name

1. Click the chart.

2. On the **Layout** tab, in the **Properties** group, click the **Chart Name** textbox.

3. Type a new chart name.

4. Press ENTER.

Change Chart Layout

1. Click anywhere in the chart.

2. Go to the **Chart Tools**, the **Design** group

3. In the **Chart Layouts**, click the chart layout that you want to use. To see all available layouts, click **More**

Change Chart Style

1. Click anywhere in the chart.

2. On the **Design** tab, in the **Chart Styles** group, click the chart style that you want to use. To see all predefined chart styles, click **More**.

Chart or Axis Titles

To make each chart easier to understand, you can add titles, such as a chart and axis titles.

To add chart title:

1. Click anywhere in the chart.
2. On the **Layout** tab, in the **Labels** group, click **Chart Title**.
3. Click **Centered Overlay Title** or **Above Chart**.
4. In the **Chart Title** text box that appears in the chart, type the text that you want.
5. To remove a chart title, click **Chart Title**, and then click **None**.

NOTE: You can also use the formatting buttons on the ribbon (**Home** tab, **Font** group). To format the whole title, you can right-click it, click **Format Chart Title**, and then select the formatting option that you want.

To add axis titles:

1. Click anywhere in the chart.
2. On the **Layout** tab, in the **Labels** group, click **Axis Titles**.
3. Do one or more of the following:
 - a. To add a title to a primary horizontal (category) axis, click **Primary Horizontal Axis Title**, and then click the option that you want.
 - b. To add a title to a primary vertical (value) axis, click **Primary Vertical Axis Title**, and then click the option that you want.
4. In the **Axis Title** text box that appears in the chart, type the text that you want.
5. To remove an axis title, click **Axis Title**, click the type of axis title to remove, and then click **None**.

Data Labels

1. On a chart, do one of the following:
 - a. Click on the **chart area** to add a data label to all data points of all data series
 - b. Click in the **data series** to add a data label to all data points of a data series
 - c. Click on a specific **data point** to add a data label to a single data point in a data series
2. On the **Layout** tab, in the **Labels** group, click **Data Labels**, and then click the display option that you want.
3. Text boxes will appear in the area of your chart based on your selection.
4. Click on the text box to modify the text.
5. To remove data labels, click **Data Labels**, and then click **None**.

NOTE: Depending on the chart type that you used, different data label options will be available.

Legend

When you create a chart, the legend appears, but you can hide the legend or change its location after you create the chart.

1. Click the chart in which you want to show or hide a legend.
2. On the **Layout** tab, in the **Labels** group, click **Legend**.
3. Do one of the following:
 - a. To hide the legend, click **None**.
 - b. To display a legend, click the display option that you want.
 - c. For additional options, click **More Legend Options**, and then select the display option that you want.

NOTE: To quickly remove a legend or a legend entry from a chart, you can select it, and then press **DELETE**. You can also right-click the legend or a legend entry, and then click **Delete**.

Move or Resize Chart

You can move a chart to any location on a worksheet or to a new or existing worksheet. You can also change the size of the chart for a better fit.

To move a chart, drag it to the location that you want.

To resize a chart, click on one of the edges and drag toward the center.

Advanced Spreadsheet Modification

Once you have created a basic spreadsheet, there are numerous things you can do to make working with your data easier. Some of these elements are hiding, freezing and splitting rows. You can also sort and filter data, these features are quite helpful when working with a large amount of data.

Hide or Display Rows and Columns

You can hide a row or column by using the **Hide** command or when you change its row height or column width to 0 (zero). You can display either again by using the **Unhide** command. You can either unhide specific rows and columns, or you can unhide all hidden rows and columns at the same time. The first row or column of the worksheet is tricky to unhide, but it can be done.

Hide Rows or Columns

1. Select the rows or columns that you want to hide.
2. On the **Home** tab, in the **Cells** group, click **Format**.
3. Under **Visibility**, point to **Hide & Unhide**, and then click **Hide Rows** or **Hide Columns**.

NOTE: You can also right-click a row or column (or a selection of multiple rows or columns), and then click **Hide**.

Unhide Rows or Columns

1. Select the rows, columns or entire sheet to unhide.
2. On the **Home** tab, in the **Cells** group, click **Format**.
3. Under **Visibility**, point to **Hide & Unhide**, and then click **Unhide Rows** or **Unhide Columns**.

TIP You can also right-click the selection of visible rows and columns surrounding the hidden rows and columns, and then click **Unhide**.

Freezing/Splitting Rows and Columns

To keep an area of a worksheet visible while you scroll to another area of the worksheet, you can either lock specific rows or columns in one area by freezing panes.

Freezing vs. splitting

When you freeze panes, Microsoft Excel keeps specific rows or columns visible when you scroll in the worksheet. For example, if the first row in your spreadsheet contains labels, you might freeze that row to make sure that the column labels remain visible as you scroll down in your spreadsheet. A solid line indicates that the row is frozen to keep column labels in place when you scroll.

When you split panes, Excel creates either two or four separate worksheet areas that you can scroll within, while rows or columns in the non-scrolled areas remain visible. This worksheet has been split into four areas. Notice that each area contains a separate view of the same data. Splitting panes is useful when you want to see different parts of a large spreadsheet at the same time.

NOTE: You cannot split panes and freeze panes at the same time.

Freeze Panes

1. On the worksheet, select the row or column that you want to keep visible when you scroll.
2. On the **View** tab, in the **Window** group, click the arrow below **Freeze Panes**.
3. Then do one of the following:

- To lock one row only, click **Freeze Top Row**.
- To lock one column only, click **Freeze First Column**.
- To lock more than one row or column, or to lock both rows and columns at the same time, click **Freeze Panes**.

NOTE: You can freeze rows at the top and columns on the left side of the worksheet only. You cannot freeze rows and columns in the middle of the worksheet.

Unfreeze panes

1. On the **View** tab, in the **Window** group, click the arrow below **Freeze Panes**.
2. Click **Unfreeze Panes**.

Split Panes

1. To split panes, point to the split box at the top of the vertical scrollbar or at the right end of the horizontal scrollbar.
2. When the pointer changes to a split pointer or, drag the split box down or to the left to the position that you want.
3. To remove the split, double-click any part of the split bar that divides the panes.

Moving or Copying Worksheets

Sometimes you may need to copy an entire worksheet instead of copying and pasting the data, which may or may not paste properly, you can use the steps below to achieve a much better result.

Move or Copy Worksheets

1. Select the worksheets that you want to move or copy.
2. On the **Home** tab, in the **Cells** group, click **Format**, and then under **Organize Sheets**, click **Move or Copy Sheet**. You can also right-click a selected sheet tab, and then click **Move or Copy**.
3. A **Move or Copy** dialog box will appear.
4. To move a sheet, in the **Before sheet** list:
 - Click the sheet that you want to insert the moved or copied sheets directly in front of.
 - Click **move to end** to insert the moved or copied sheets after the last sheet in the workbook and before the **Insert Worksheet** tab.
5. To copy the sheets, in the **Move or Copy** dialog box, select the **Create a copy** checkbox.

NOTE: When you create a copy of the worksheet, the worksheet is duplicated in the workbook, and the sheet name indicates that it is a copy—for example, the first copy that you make of Sheet1 is named Sheet1(2).

NOTE: To move sheets in the current workbook, you can drag these selected sheets along the row of sheet tabs. To copy the sheets, hold down CTRL, and then drag the sheets; release the mouse button before you release the CTRL key.

Move or Copy to a Different Workbook

1. In the workbook that contains the sheets that you want to move or copy, select the sheets.
2. On the **Home** tab, in the **Cells** group, click **Format**, and then under **Organize Sheets**, click **Move or Copy Sheet**.
3. In the **Move or Copy** dialog box, click the dropdown list in the **To book** box, then:
 - Click the workbook to which you want to move or copy the selected sheets.
 - Click **new book** to move or copy the selected sheets to a new workbook.
6. To move a sheet, in the **Before sheet** list:
 - Click the sheet that you want to insert the moved or copied sheets directly in front of.
 - Click **move to end** to insert the moved or copied sheets after the last sheet in the workbook and before the **Insert Worksheet** tab.
7. To copy the sheets, in the **Move or Copy** dialog box, select the **Create a copy** checkbox.

Sorting Data

Sorting data is an integral part of data analysis. You might want to arrange a list of names in alphabetical order, compile a list of product inventory levels from highest to lowest, or order rows by color icons. Sorting data helps you quickly visualize and understand your data better, organize and find the data that you want, and ultimately make more effective decisions.

Sort Data in Single Column

1. Select a column of data in a range of cells.
2. On the **Data** tab, in the **Sort & Filter** group, do one of the following:
 - To sort in ascending or smallest to largest order, click **Sort A to Z**.
 - To sort in descending or largest to smallest order, click **Sort Z to A**.
3. To reapply a sort after you change the data, click a cell in the range or table and then, on the **Data** tab, in the **Sort & Filter** group, click **Reapply**.

Sort Data in Multiple Columns or Rows

You may want to sort by more than one column now when you have data that you want to group by the same value in one column now, and then sort another column now within that group of equal values.

1. Select a range of cells with two or more columns of data.
2. On the **Data** tab, in the **Sort & Filter** group, click **Sort**.
3. The **Sort** dialog box will appear.
4. Under **Column**, in the **Sort by** box, select the first column that you want to sort.
5. Under **Sort On**, select the type of sort.
 - To sort by text, number, or date and time, select **Values**.
 - To sort by format, select **Cell Color**, **Font Color**, or **Cell Icon**.
6. Under **Order**, select how you want to sort.
 - For text values, select **A to Z** or **Z to A**.
 - For number values, select **Smallest to Largest** or **Largest to Smallest**.
 - For date or time values, select **Oldest to Newest** or **Newest to Oldest**.
 - To sort based on a custom list, select **Custom List**.
7. To add another column to sort by, click **Add Level**, and then repeat steps four through six.
8. To copy a column to sort by, select the entry and then click **Copy Level**.
9. To delete a column to sort by, select the entry and then click **Delete Level**.
10. To change the order in which the columns are sorted, select an entry and then click the **Up** or **Down** arrow to change the order.
11. To reapply a sort after you change the data, click a cell in the range or table and then, on the **Data** tab, in the **Sort & Filter** group, click **Reapply**.

Finalizing a Spreadsheet

To complete your spreadsheet there are a few steps to take to ensure your document is finalized.

Using the "Spell Check" Feature

Excel does not have the same spell check features as Word and PowerPoint. To complete a Spelling and Grammar check, you need to use the Spelling and Grammar feature.

1. Click on the **Review** tab
2. Click on the **Spelling & Grammar** command (a blue checkmark with ABC above it).
3. A **Spelling and Grammar** box will appear, correct any Spelling or Grammar issue with the help of the box.

Save a Spreadsheet

To save a document in the format used by Excel 2010 and Excel 2007, do the following:

1. Click the **File** tab.

2. Click **Save As**.
3. In the **Filename** box, enter a name for your document.
4. Click **Save**.

To save a document so that it is compatible with Excel 2003 or earlier, do the following:

1. Click the **File** tab.
2. Click **Save As**.
3. In the **Save as type** list, click **Excel 97-2003 Document**. This changes the file format to .xls.
4. In the **Filename** box, type a name for the document.
5. Click **Save**.

Help

If you need additional assistance when completing your document you can use the help feature.

1. Click on the blue circle with the white question mark command
2. A **Help** box will appear.
3. Click in the **Search Help** text box and type what you need help with
4. Click the magnifying glass next to the text box and the possible solutions will appear.

Lab

Creating and Designing presentation in Microsoft PowerPoint

Objective:

Creating and Designing presentation in Microsoft PowerPoint.

Microsoft PowerPoint:

PowerPoint is a system in the Microsoft Office Suite that enables you to present information in office meetings, lectures and seminars to create maximum impact in a minimal amount of time. PowerPoint presentations can amplify your message, accelerate the information being absorbed and assist with comprehension enabling faster decision making.

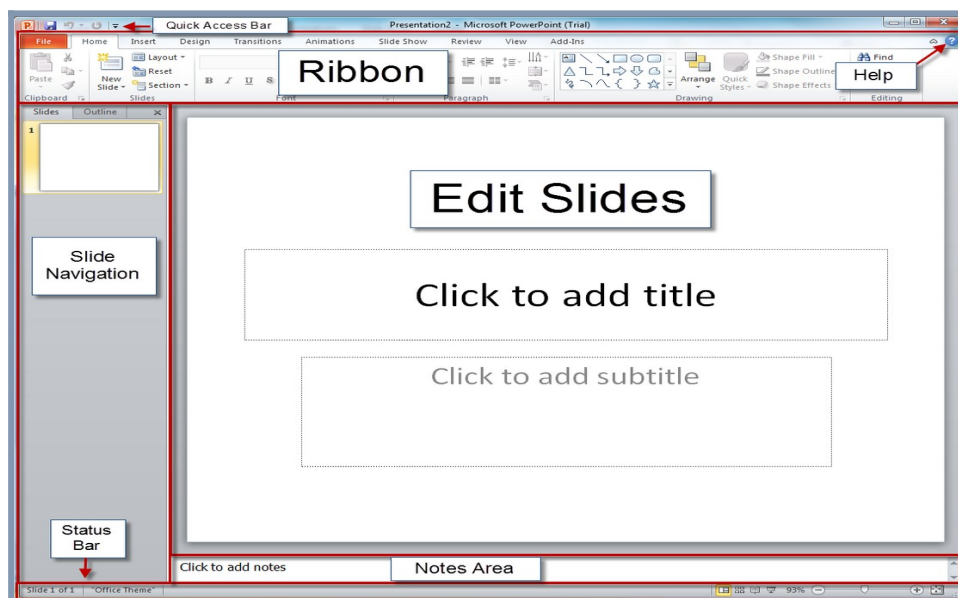
File menu:

The File menu contains commands such as Save, Open and Print.

Quick Access Toolbar:

The Quick Access Toolbar contains commonly used tools and is located at the top or bottom of the ribbon. You can easily add more tools from the Quick Access menu or by right-clicking on a tool and selecting Add to Quick Access toolbar.

Working With Presentation:



New PowerPoint Document:

When you first open PowerPoint you will see what's called the **Normal** view. The **slide pane** is the big area in the middle. This is the area you will work in to create your slides. On each slide, you will see various boxes with the dotted borders which are called **placeholders**. This is where you type your text. Placeholders can be customized to different sizes and can contain pictures, charts, and other non-text items. On the left of the screen are **thumbnail** versions of the slides in your presentation; the slide you're reworking will be highlighted. The bottom area is the **notes pane**, this is where you type speaker notes that you can refer to when you present. You can also print speaker notes to use when presenting a slide show.

Saving A Presentation:

You have created your first presentation. You now need to save this presentation to your hard disk. To do this, click on the **Save** icon towards the top-left of your screen. You will see the **Save As** dialog box displayed. Write the filename in the text box & click on **save** button.

Opening A Presentation:

You can open a presentation by pressing (ctrl+o). Or To open a file, click on the **File Tab**, You will see the **Open** dialog box displayed. Normally it will display the contents of a folder called **Documents** or **My Documents**. Now select that file which you want to open, To open the file either click on the **Open** button at the bottom-right of the **Open** dialog box or double-click on the filename.

Closing A Presentation:

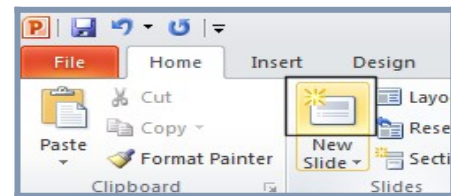
To close the presentation click on the **File Tab** (top-left of your screen) and then click on the **Close** command. The PowerPoint window will still open, but no presentations are open within the program.

Creating A Presentation:

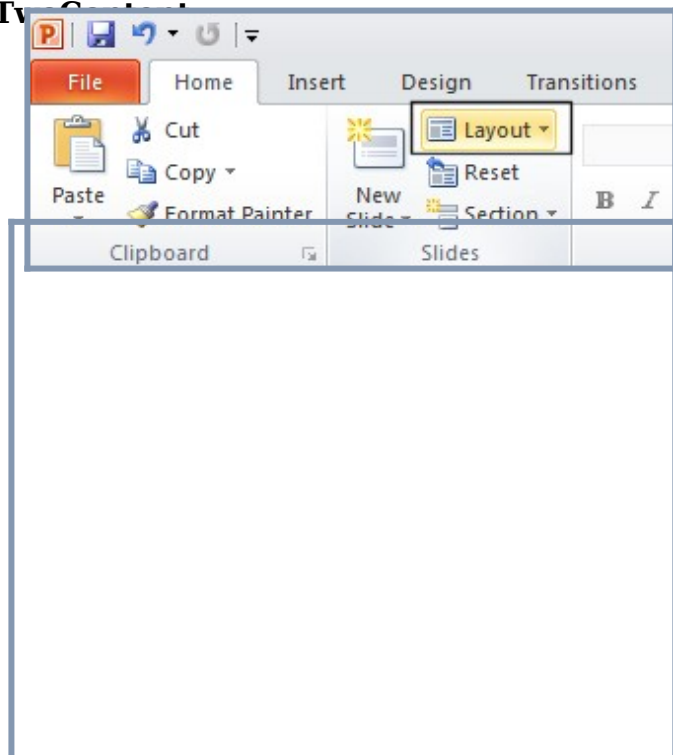
Adding Or Inserting New Slide:

You can add new slides to the presentation by pressing (ctrl+M) and To do this, if necessary, click on the **Home** tab, and you will see the **New Slide** button displayed in the Ribbon.

- Within the **Home** tab, click on the **Layout** button.



- From the dropdown options select **Two Content**.



Undo and Redo:

When creating or editing a PowerPoint presentation, always remember that if you make a mistake, in most cases you can click on the Undo icon and undo the mistake. The Undo icon is displayed at the top-left of your screen. Try selecting a word within a slide and then pressing the Del key to delete it. Then click on the Undo icon to undo the deletion.

Formatting Text

Many of your slides will require you to enter text in the placeholder boxes.

- To change the text font, color, and size, use commands in the Font group.
- To change paragraph formatting such as bullet type, text indentation, and line spacing, use the commands in the Paragraph group.

Adding a Design Theme

A theme includes a background design, color scheme, font types, font sizes, and placeholder positions in one package. Every new presentation starts out with the default theme, called Office theme, which is a white background and black text. However, you can change the theme to a wide variety of options. To find and apply a theme, click the Design tab on the Ribbon.

- When you point and hover on any theme thumbnail, a preview of the theme will appear on the slide. To apply the theme to your slides, click on the thumbnail design you like.



NOTE:

At the time a theme is selected, it can be changed at any time during the creation of your slides. However, the theme can alter the position of placeholders, so your text may be automatically adjusted to properly fit the placeholder of the theme, causing unexpected formatting changes to your slides.

Lab

Understanding insertion of ClipART, Charts, textbox inside a presentation slide.

Objective:

Understanding insertion of ClipART, Charts, textbox inside a presentation slide.

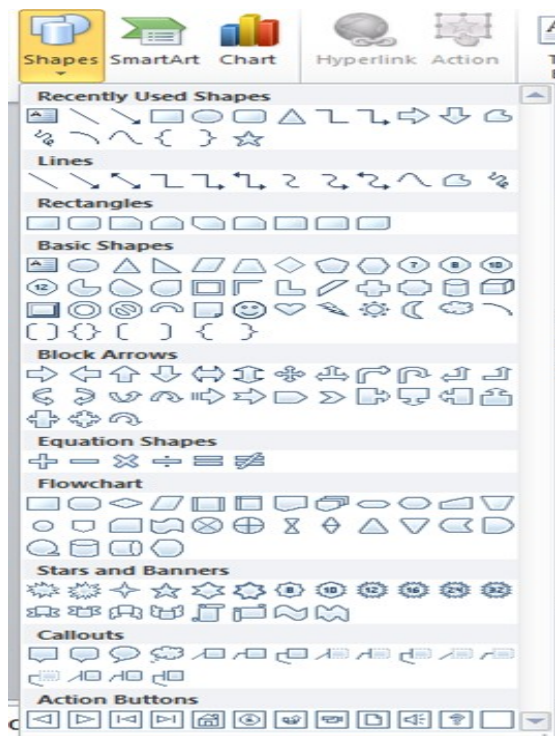
Inserting ClipArt

If you would like to add an additional dimension to a PowerPoint slide, you can add ClipArt to your slides. ClipArt includes pictures, sounds, and videos. There are two ways to initiate inserting ClipArt depending on where you would like the graphic to be located.

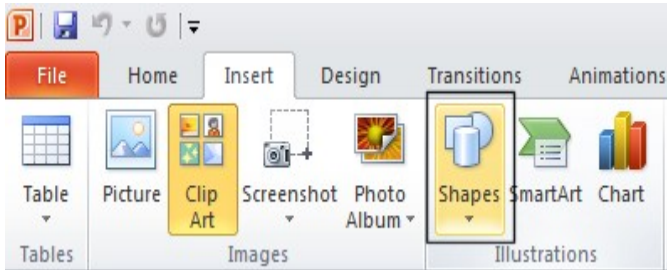
- The first method is to go to the Insert Ribbon and click on the ClipArt command. You can also click on the Audio or Video commands and opt to pick from the ClipArt gallery. The second method is to click on the ClipArt icon in a placeholder.
- The ClipArt task pane will then open on the right. Type a keyword in the Search for box that suggests the type of clips you may want. Use the Results should be dropdown to select the media type to search in, then click Go.
- Clips that fit the keyword will appear in the box below. Click on the clip that you would like on your slide and it will be automatically appear.

Inserting Shapes

- Display the slider relating to the mouse.
- Click on the **Shapes** icon (contained within the **Illustrations** section of the **Insert** tab).



- This will display a dropdown from which you can select the required shape.



Inserting text into a shape

Select the shape and click on the shape using the right mouse button. From the popup menu displayed select the Edit Text command.

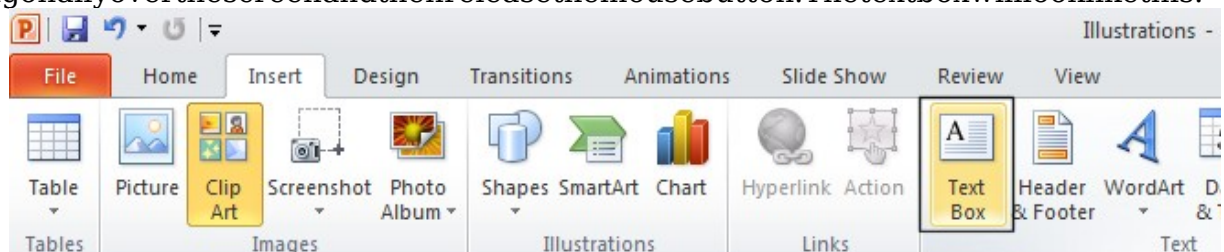
NOTE: You can also just select the shape and start typing.

- If you look closely you will notice that the text insertion point is now within the shape.

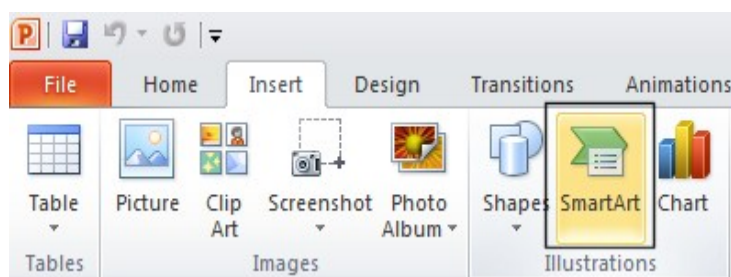
Inserting a textbox

Insert a new **Blank** slide, by clicking in the lower part of the **New Slide** icon (within the **Slide** section of the **Home** tab).

- Click on the **Text Box** icon within the **Text** section of the **Insert** tab.
- The mouse pointer changes shape. Move the mouse pointer to the position within the slide that you want to start the textbox. Press the mouse button and keep it pressed. Move the mouse pointer diagonally over the screen and then release the mouse button. The textbox will look like this.



- Type your name into the textbox.
- Save your changes.



Inserting SmartArt

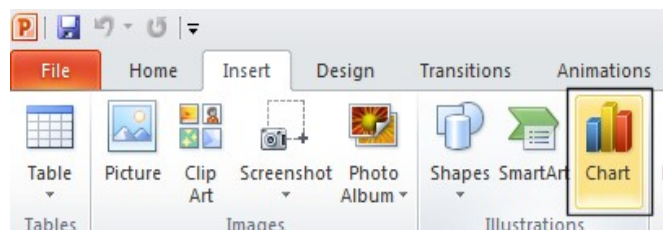
Insert a new **Blank** slide, by clicking in the lower part of the **New Slide** icon (within the **Slide** section of the **Home** tab).

- Click on the **SmartArt** icon (contained within the **Illustrations** section of the **Insert** tab). This will display the **Choose a SmartArt Graphic** dialog box.
- Scroll down through the **List** section and you will see a fantastic range of SmartArt graphics that you can insert into your presentation.

Charts

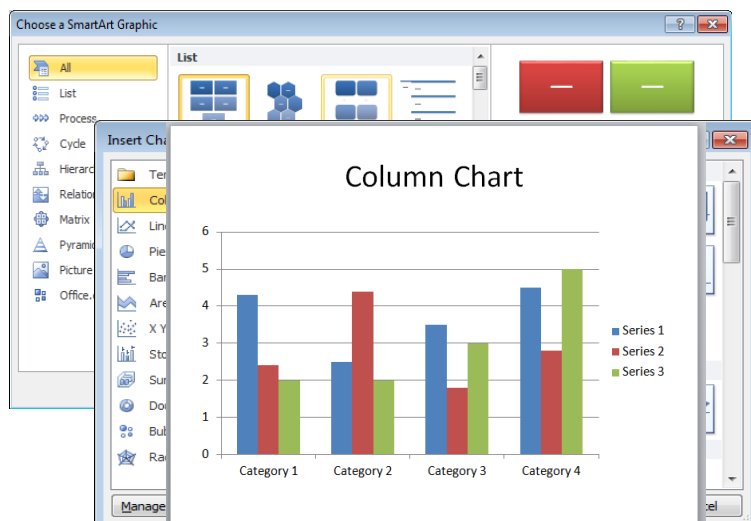
Inserting Charts

- Open a presentation called **Charts**.
- Click on the **Chart** icon (contained within the **Illustrations** group under the **Insert** tab).
- This will display the **Insert Chart** dialog box.
- You can select a chart type such as **Column**, **Line** or **Pie**. You can then select a specific format in a option for these selected chart type. In this case select a **Column** chart and from the right side of the dialog box, and then select a **Clustered Column**.
- Click on the **OK** button. Your screen will now change.

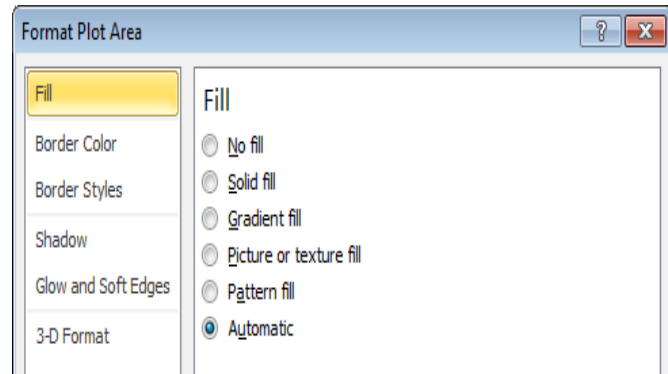


Changing the background color in the chart

- Click on the chart to select it.



- Rightclickoveranemptypartofthechartbackgroundandyouwillseeapopupmenudisplayed.
- Clickonthe**FormatPlotArea**command.Youwillseethe**FormatPlotArea**dialogboxdisplayed.
- Tosetasolidcolorbackgroundforyourchartclickonthe**SolidFill**button.Thenclickonthe**down arrow**nexttothe**Color**sectionandselectacolorfromthedropdownthatisplayed.
- Clickonthe**Close**buttonwithinthedialogbox andyouwillseethecolorappliedtoyourchart background,asillustrated.



Changing the column, bar, line or pie slice colors in the chart

- Youcanapplycolorstotheelementswithinyourchart.Clickononeofthecolumnswithinthechart andyouwillnoticethattheothersimilarcolumnsarealsoselected.
- Rightclickoveroneoftheselectedcolumnsandyouwillseeapopupmenudisplayed.Selectthe**FormatDataSeries**command.
- Youwillseethe**FormatDataSeries**dialogboxdisplayed.
- Touseasolidcolortofillthecolumn,clickthe**Fill**sidetabandthenalsoclickonthe**SolidFill**button.
- Clickonthe**downarrow**nextinthe**Color**sectionandfromthedropdowndisplayed,selectacolor.
- Clickonthe**Close**buttonandthecolorwillbeappliedtotheselectedcolumns.Intheexampleshownwehaveappliedthecolor**Black**.Trythisnow.

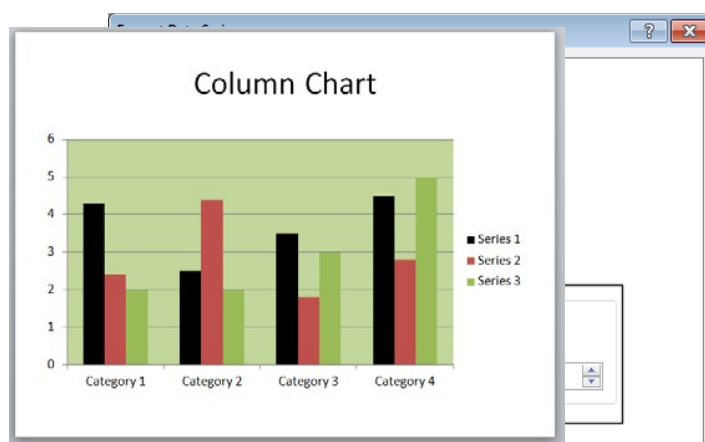
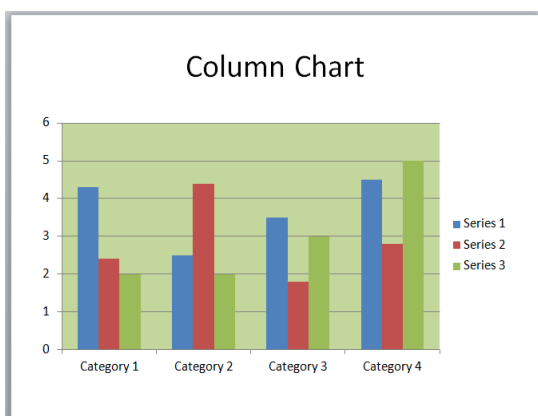


Table:

Creating a table

- Click on the **Insert Table** icon displayed in the centre of the slide.

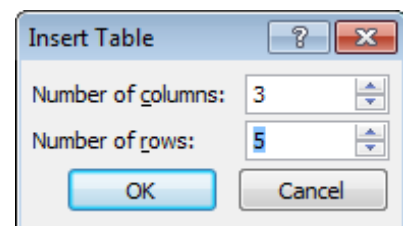
TIP: Alternatively, to insert a table click on the **Insert** tab and click on the **Table** icon displayed within the Ribbon.

- You will see the **Insert Table** dialog box as illustrated.
- Use the dialog box to create **3 columns** and **5 rows**, as illustrated.
- Click on the **OK** button and the table will be inserted into the slide.

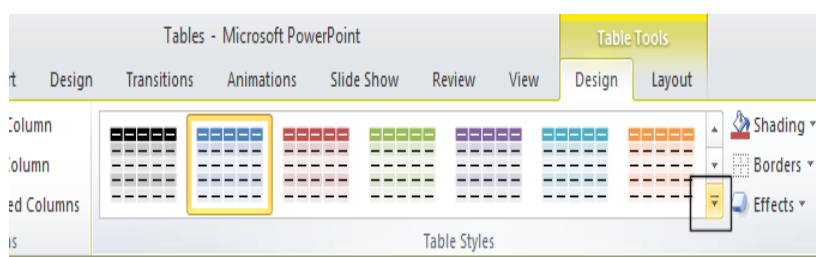
Applying a style to a table

If necessary, double click within the table and you should see the **Table** Ribbon displayed.

- Click on the **down arrow** (called the **More** arrow) at the right of the **Table Styles** section.
- You will see a dropdown displaying a range of different styles. Click on one and see the effect on your table.



Deleting rows or columns



Click within a cell, within the column that you wish to delete. Right click within the cell and from the popup menu displayed, select the **Delete Columns** command.

To delete a row:

Click within a cell, within the row that you wish to delete. Right click within the cell and from the popup menu displayed, select the **Delete Rows** command.

Use the **Undo** icon to bring back the column and row you just deleted.

Inserting rows or columns

- Right click over the selected cell and from the popup menu displayed select the **Insert** command. From the sub menu displayed select **Insert Column to the Right**. You will see the new column displayed to the right of the selected cell.
- To insert a row right click over the selected cell and from the popup menu displayed select the **Insert** command. From the sub menu displayed select **Insert Rows Above**. You will see the row inserted as illustrated. Save your changes

Lab

Understanding slideshow techniques, and animation Styles in MS Power Point

Objective:

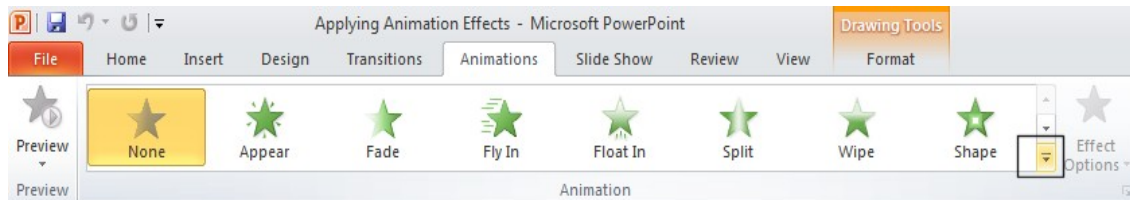
Understanding slideshow techniques, and animation Styles in MS Power Point

Slide Shows

Adding slideshow animation effects

Open a presentation

- You can add animation effects to the text and graphics within your slides.
- Select the **title** area on the first slide.
- Click on the **Animations** tab and within the **Animation** group click on the **down arrow** to the right of the group (i.e. on the **More** button).

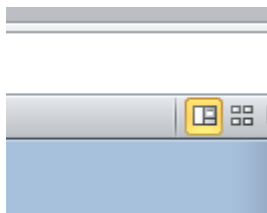


- Select an animation effect, such as **Fly In**. You will notice that the **title** area is marked with an animation number 1.
- Select the sub-title area, select an animation effect, such as **Bounce**. You will notice that the slide is marked with two animation numbers, as illustrated, one for the **title**, which is marked with a 1, indicating that this is the first animation, while the **sub-title** is marked with a 2, indicating that this is the second animation.
- Click on the **Slide Show** button, at the bottom-right of the screen.
- Click the mouse button once, and you will see an animated **title** appear.
- Click the mouse button again and an animated sub-title will be displayed.
- Click the mouse button again to see the animated graphic appear.
- Click on more time and the second slide, which does not contain any animations, will be displayed. Press **Esc** to escape from the slide show.

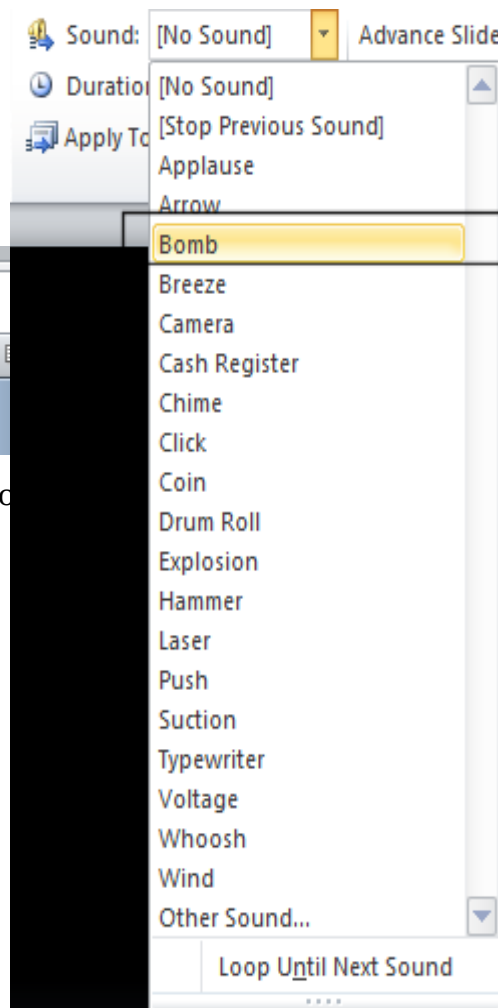
Slide show transitions and timings

- Click on the **Transition** tab. You will see a selection of transitions displayed within the ribbon.
- Click on the **down arrow** to the right of the **Transition** group and you will see more transitions displayed in a pop-down menu. Select the **Cube** transition.

•



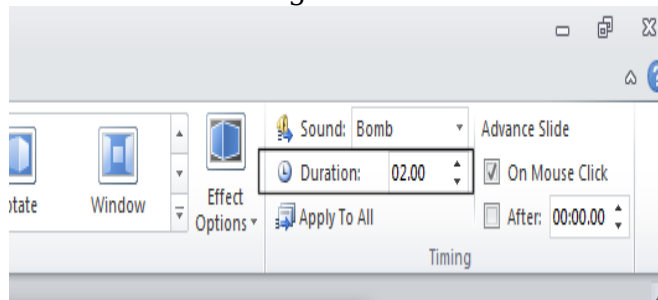
You can add sound to the slide animation. To do this click on the **Sound** button. You will see a drop-down list displayed. Click on the **Bomb** sound.



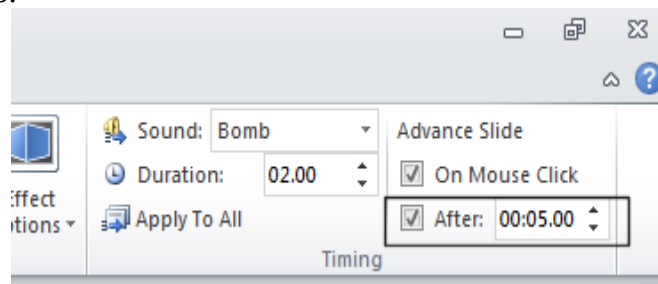
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- You can control the duration of the slide animation. To do this, click on the **up arrow** to the right of the transitions **Duration** button and change the value to **2.00**.



- You can set how the slide will advance. Click on the checkbox next to the **After** option and enter a value of **5.00** seconds.



- You can apply the slide animation to all the slides by clicking on the **Apply to All** button.
- Run the slideshow and the transition effects will display as you configured them to run. Save your changes.

Using Find

Click the **Find** button, located in the **Editing** group under the **Home** tab.

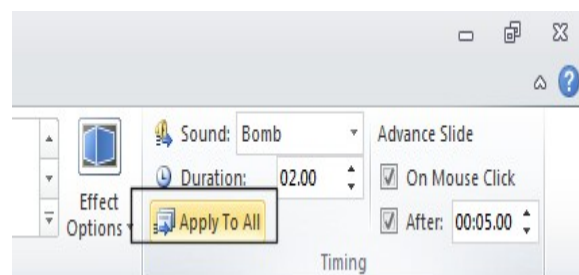
TIP: Use the **Ctrl+F** keyboard shortcut to start the **Find** function.

- The **Find** dialog box will be displayed. Type the word into the **Find what** box.
- Click on the **Find Next** button. PowerPoint will locate and select the first instance of the word in the presentation. Click on the **Find Next** button repeatedly to find all occurrences of the word.
- When you reach the end of the search, PowerPoint will display the following dialog box.
- Click on the **OK** button.

Using Replace

Click the **Replace** button, located under the **Editing** group of the **Home** tab.

TIP: Use the **Ctrl+H** keyboard shortcut to start the **Replace** function.



tin
the

- The **Replace** dialog box will be displayed. Type the word into the **Find what** box.
- Type the word **PC** into the **Replace with** box.

- Click on the **Find Next** button. PowerPoint will locate and select the first instance of the word computer in the presentation. To replace that instance of the word computer with PC click on the **Replace** button. Repeat this process to replace all occurrences of the word computer with PC.
- When you reach the end of the search PowerPoint will display the following dialog box. Click on the **OK** button.

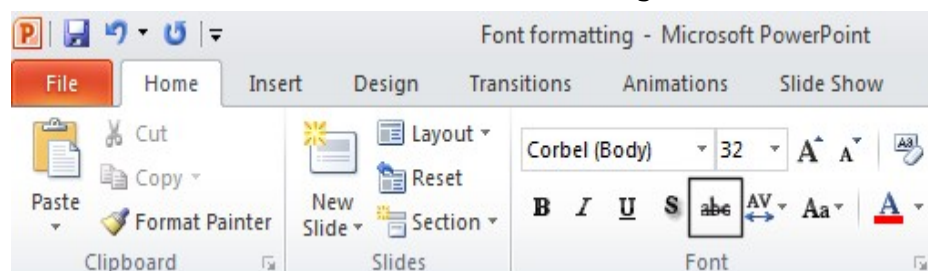
Strikethrough effects

Select a word and experiment with applying strikethrough formatting using the **Strikethrough** icon.

Shadow effects

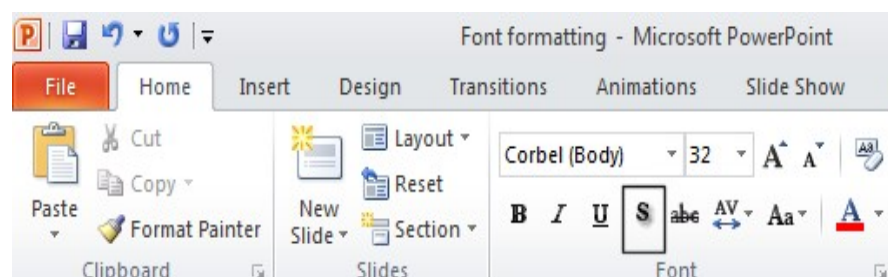
Select a word within the last paragraph and experiment with applying **Shadow** formatting. You may find this effect is easier to see when combined with **Bold** formatting.

Character spacing

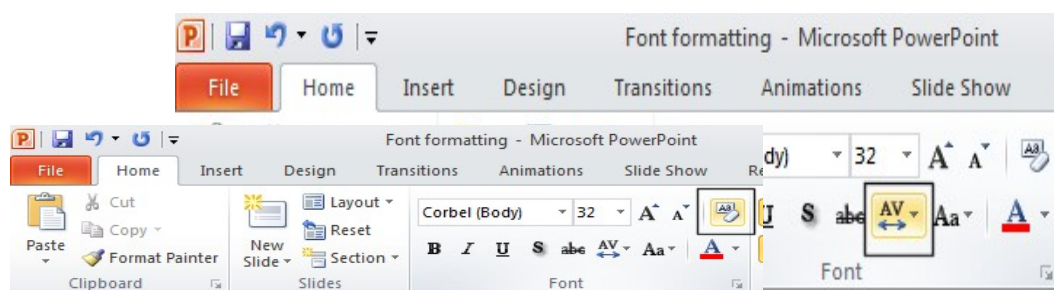


Display the slider relating to the CPU.

- Select the first paragraph.
- Click on the **Character Spacing** icon. This will display a dropdown from which you select the required type of character spacing. Experiment with applying different types of character spacing and observe the effects. Reset the character spacing back to 'Normal' before continuing.



Clearing all text formatting



Select a paragraph of text that you have previously formatted. Click on the **Clear All Formatting** icon and the formatting will be removed.

PowerPoint Presenter Notes:

If you look carefully at your screen you will see a section towards the bottom displaying the message '**Click to add notes**'. You can use this area to add presenter notes to a presentation. These can be used as memory aids and prompts by the presenter, but are not designed to be seen by persons viewing the presentation. Add the following notes to the title page.

NOTE: To enlarge or decrease the notes area, place your cursor on the border and click and drag the cursor.

Text direction

- Click on the word you want to change direction.
- Click on the **Home** tab and within the **Paragraph** group click on the **Text Direction** icon.
- This will display a dropdown list of text direction options. Select the **Rotate all text 90°** option.

Spell-checking

- Open a document called **Spellcheck**. This presentation contains a number of spelling errors. To spell check the document, click on the **Spelling** icon located within the **Review** tab.

TIP: The keyboard shortcut to run the spell checker is **F7**.

- Follow the on-screen prompts to correct the errors within the document.

Assignment:

Make 5 powerpoint slides with different background, about your bio-data, about computer engineering, about DUET, about computer computation, about powerpoints simultaneously.

Date: _____

Lab#10INTRODUCTIONTOPROGRAMMINGWITHTHEDEV++IDE.

OBJECTIVES:

- O1: TounderstandthecompilerofclanguageDEV++.
- O2: Howto create, compile and execute program.

EQUIPMENTTOBEUSEDINTHELAB

- Acomputersystem
- DEV++installed

THEORY:

IntroductiontotheIDE:

Dev-C++isafull-featuredIntegratedDevelopmentEnvironment(IDE)fortheC/C++programminglanguage.AssimilarIDEs,itofferstotheprogrammerasimpleandunifiedtooltoedit,compile,link,anddebugprograms.Italsoprovidessupportforthemanagementofthefiles ofaprogramin“projects”contain ingalltheelementsrequiredtoproduceafinalexecutableprogram.

DocumentationabouttheCprogramminglanguageandtheIDEitselfcanbeaccessedfromDev-C++.

First steps

The application development process encompasses the following steps:

1. Create project

The type of application and the programming language to be used are specified.

2. Write source code

Write the program in C and save the source code file.

3. Compile and link the code

The source code is compiled and linked to generate a running program. Other files of the project may be created.

4. Fix compilation errors, if any

If the syntax of the program is not correct, the compilation fails and the compiler generates a message related to the error/s. The programmer must correct the errors.

5. Run the program

Run the program to validate the functioning.

6. Fix execution errors, if any

If the actions performed by the program are not as expected, it is necessary to correct the source code. It may be also convenient to use the debugger to find complex errors.

After indicating the folder where the project configuration file (.dev) will be saved, the IDE generates a basic source code file (by default, main.c). These files are not saved in the project folder until the programmer saves or compiles the program (see below).

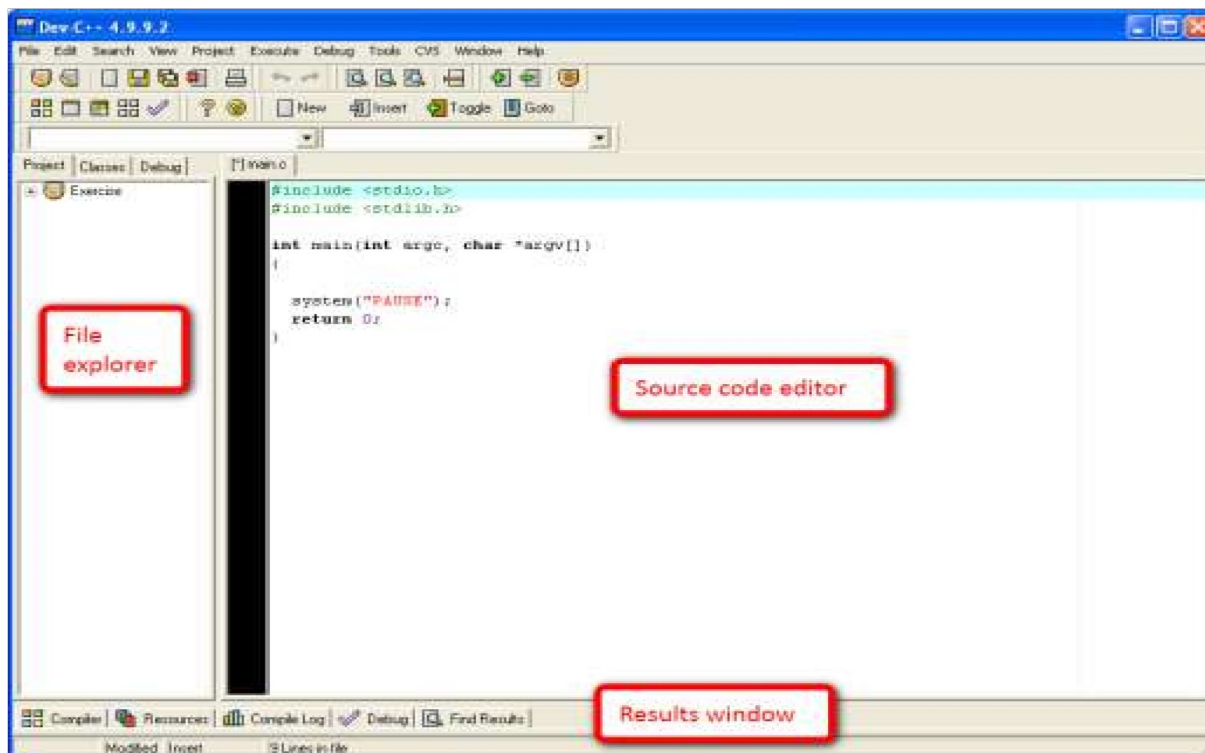


Figure 7.1. Dev-C++ launches after project creation

The IDE window includes three sub-windows: the Project Files Explorer, the Result Tabs, and the Source Code Editor. These windows can be resized and minimized.

The Files Explorer window shows the name of the project and the included files. The Project tab usually contains a single file with the source code of the program. In this pane, we can find two additional tabs: Classes and Debug. Classes tab shows the functions of the program. Debug tab shows watched variables in the debugging process.

The Results window is used to present the results of the actions of the IDE: compilation errors, compiling directives, debugging commands, etc. The Source code editor shows the code of the program.

Files created:

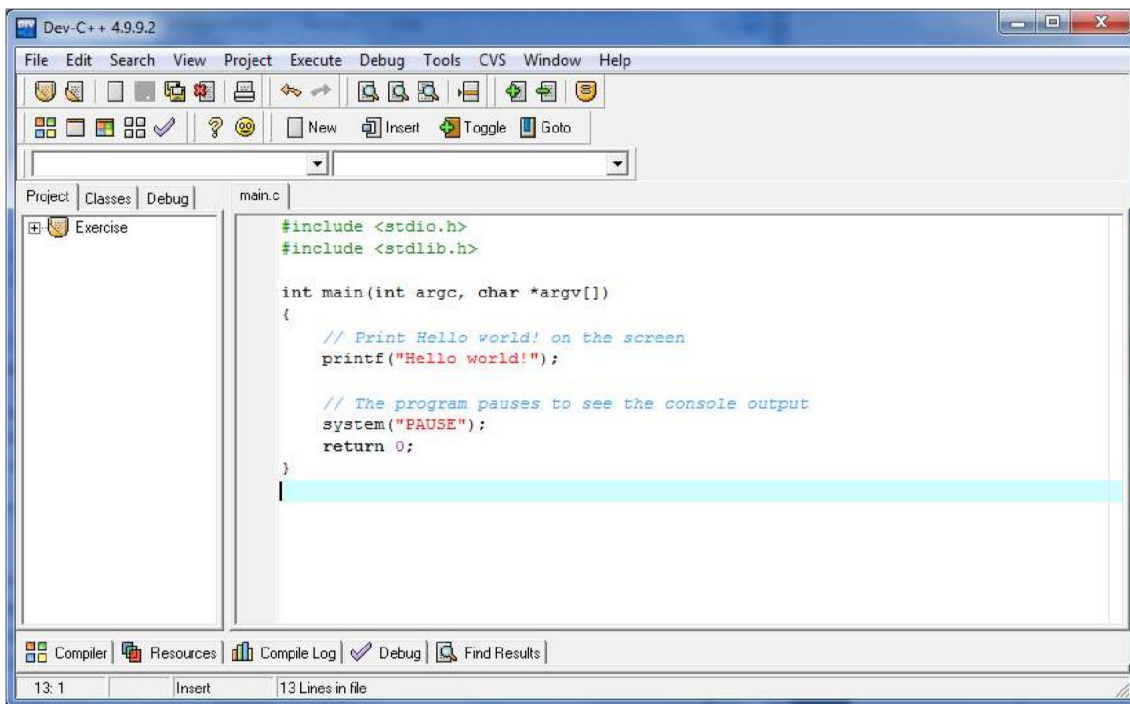
The files of the project are saved when the source code file is saved. The remaining files of the project are saved when the application is compiled and stored in the project folder.

File	Extension	Description
Project file	.dev	Project configuration data
Makefile	.win	Required for the compilation process. Manages program dependencies and includes instructions for the linker
Source code file	.c	Source code
Object file	.o	Object code resulting from the compilation of the source code. Each .c has a corresponding .o after the compilation
Executable file	.exe	Executable application

The editor highlights with different colors keywords and other elements of the C language. The classic scheme uses:

- Light blue for comments
- Green for included libraries
- Red for text strings
- Bold black for C keywords

The Hello World program is shown as follows:



CompilationAndLink:

To run a program, the source code must be compiled and linked. Dev-C++ performs the complete process by clicking the Compile button (or Ctrl+F9).

While the compilation and link process is being performed, the IDE shows a dialog with related information. If the process is successful, the window shows the message Done.

The Compile Log tab shows information about the process.

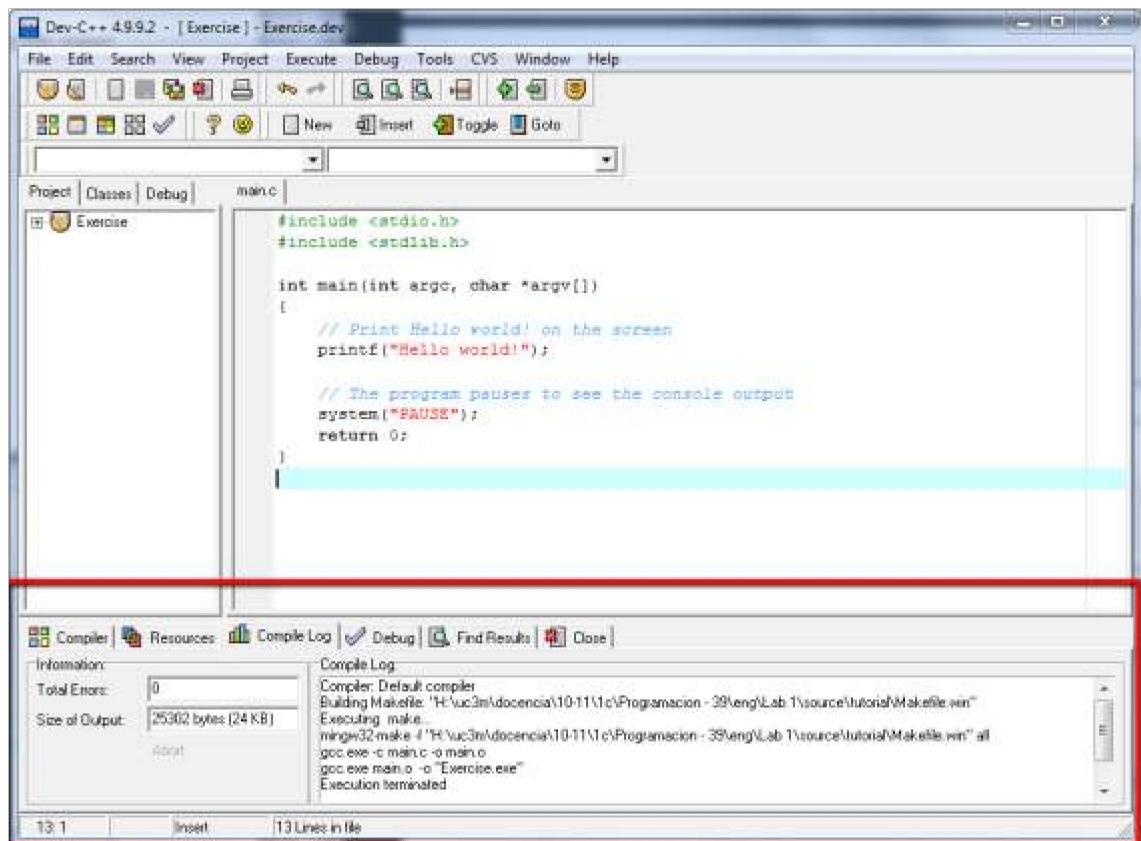


Figure 7.3 Compilation process output

Fixing compilation errors:

Compilation errors are errors that are detected by the compiler. They are also named design time errors. To correct compilation errors, it is recommended to solve the first one and then to re-compile the program again, since next errors are frequently wrong sections of the code that the compiler cannot interpret as a result of the first error.

Dev-C++

+ underlines in red the line of code where the compilation error has been detected. The Compiler tab of the Results window provides a detailed description of the error. The Compile Log shows the error message issued by the compiler program.

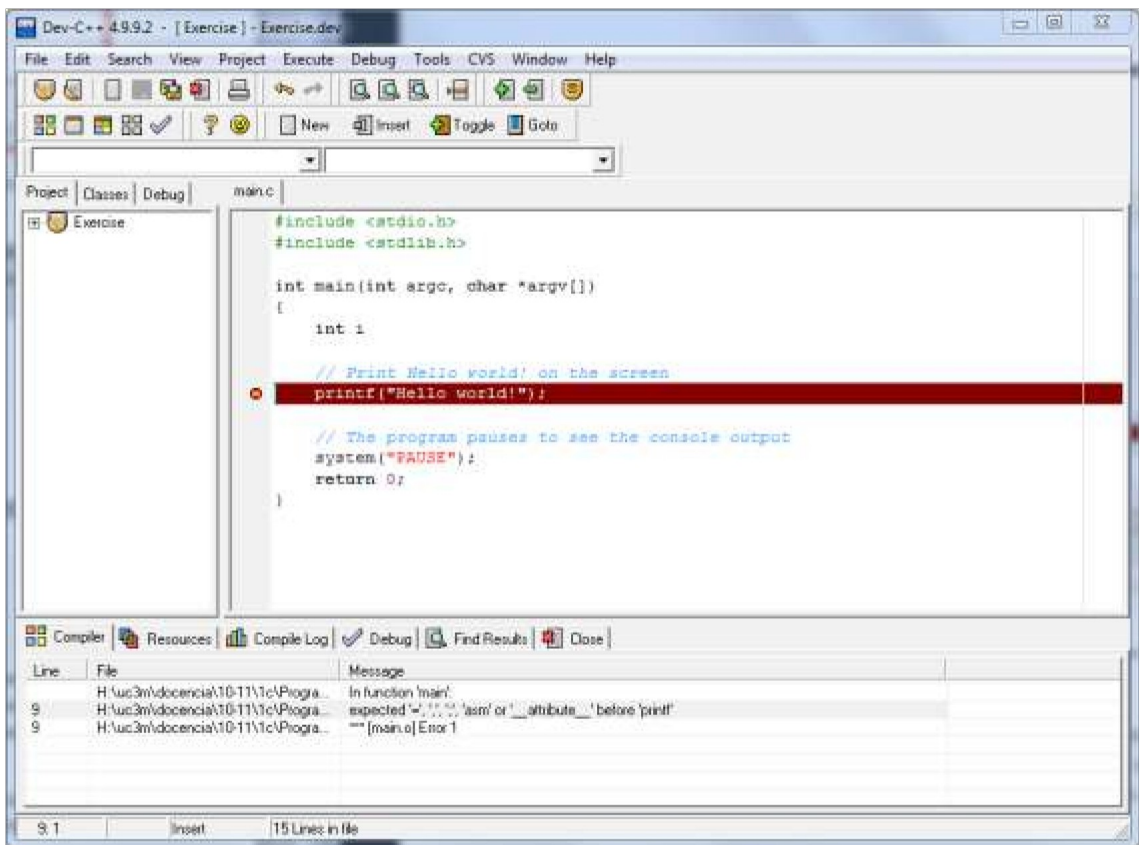


Figure7.4.Compilationerrors

Commonerrorsare:selectingawrongprojecttype(insteadofConsole),selectingawronglanguatype(instead ofC),orlibraryinclusionerrors(wrongsyntax,librarymissing,etc.).

Runningtheprogram:

Asaresultofthelinkprocess,anexecutableprogramiscreated.Torunthisprogram,wehavetoclicktheRunbutton (Ctrl+F10).ItispossibletocompileandruntheprograminasinglestepbyclickingtheCompile&Runbutton(F9).

Whentheprogramiscompiledandlinkwiththeseoptions,wecanruntheprogramindebugmodebyclickingontheDebugbutton(F8).Thismodeallowsrunningtheprogramstepbystep(instructionsareexecutedone-by-one),stoppingtheexecutionatbreakpoints,orwatchthevalueofavariableatanytimeoftheexecution.

WhentheprogramrunsinDebugmode(F8),theexecutionispausedatthisline.Fromthispoint,theexecutioncan continuenormallyorstepbystep.Thecurrentinstructionishighlightedinblue.

TheDebugtaboftheResultswindowshowsthedebuggingcommandsofDev-C++ .Asdepictedinthefollowingpicture,itispossibletocontinuetheexecution,runtheprogramstepbystep,runtothecursor,etc.



Figure 7.5. Debug mode option to run a program

Create a new program:

To create a new program, replace the previous source code with the new code. We cannot have two files with a main function in the same project. Therefore, it is convenient to create a different project for each program.

CONCLUSION:

Date: _____

**Lab#(10-A) TO STUDY BASIC BUILDING BLOCKS OF C-
LANGUAGES SUCH AS DATA TYPES AND INPUT-OUTPUT FUNCTIONS**

OBJECTIVES:

- O1: To understand the basic elements used to construct C elements.

O2:

How to create character set, identifiers, keywords, data types, constants, variables, expressions, statements and escape sequences.

EQUIPMENT TO BE USED IN THE LAB

- A computer system
- DEV C++ installed

THEORY:

Comments:

Comments statements will not be compiled. Comments are simply the statements to improve program readability and to document program properly. Comments begin with `/*` and end with `*/`, text is placed in between them. `/*Lab Session 7*/`

printf() Function: This function is used to output combination of numerical values, single character and strings.

Syntax:-

```
printf("format specifier", variable or constant); printf("text");
```

Example:-

```
printf("Area of circle is %f sqmm", 3.756);
```

scanf() Function:

The purpose of `scanf()` function is to accept data from keyboard, and place that data to a memory location specified in its argument.

Syntax:-

```
scanf("format specifiers", address of variable);
```

Escape Sequences:

These are non-printing characters. They are a special character set, each with specific meaning. An escape sequence always begins with a backslash and is followed by one or more special characters.

Escape Sequence	Meaning
\n	New Line
\t	Horizontal Tab
\a	Alert(Bell)
\\	Back Slash
\"	Quotation Mark
\f	Form feed
\r	Carriage Return
\0	Null

Table:Escape sequences

Variables:

A variable name is a location in memory where a value can be stored for use by a program. All variables must be defined with a name and a data type in the code before they can be used in a program.

A variable name in C should be a valid identifier. An identifier is a series of characters consisting of letters, digits and underscore and does not begin with a digit. C is case sensitive i.e. area and Area can't be treated as same.

There are certain reserved words called **Keywords** that have standard, predefined meanings in C. These keywords can be used only for their intended purpose; they can't be used as programmer-defined identifiers.

Data Types:

C supports several different types of data, each of which may be represented differently within the computer's memory. The basic data types are listed below in Table 2.2.

Data Type	Meaning	Bytes
char	Character	1
int	Integer	2
short	Short Integer	2
long	Long Integer	4
Unsigned	Unsigned Integer	2
Float	Floating	4
Double	Number(Decimal) Double Precision Floating Point Number	8

Table 2.2: Data type and storage allocation

Format Specifiers:

Format specifiers specify that which type of data has to be printed or read into. Following is a list of different format specifiers in table 2.3.

Specifiers	Meaning
%c	Character
%d	Integer
%f	Float value
%e	Float value in exponential form
%u	Unsigned Integer
%x	Hexadecimal integer (unsigned)
%o	Octal value
%s	String

Table 2.3: Format specifiers

LAB TASK:

1) Identify and correct the errors in the following statements.

a) `scanf("d",value);`

b) `printf("The answer of %d + %d is \n",x,y);`

c) `scanf("%d%d",&number1,number2);`

d) `printf("The number is %d/n",&number1);`

2) What do these codes print?

a) `printf("\n*\n**\n***\n****\n*****");`

b) `printf("This is base");`

c) `printf("\n\t\t1\n\t\t2\n\t\t3\n\t\t4\n\t\t5\n\t\t6\n\t\t7");`

LAB TASK:

1. Write a program which can only print your name, roll number, Department name.
2. Create a program which can integer type input from user.

LEARNING OUTCOMES

Upon successful completion of the lab, students will be able to:

LO1: Students able to take different input from user.

LO2: Able to use of **printf** and **scanf**.

Date: _____

Lab#11 TO STUDY THE DIFFERENT TYPES OF ARITHMETIC AND LOGICAL OPERATORS.

OBJECTIVES:

- O1: To understand the logical and arithmetic operators in C language.
- O2: How these operators are used in preliminary programs..

EQUIPMENT TO BE USED IN THE LAB

- A computer system
- DEV C++ installed

Theory:

In C, there are various operators, used to form expressions. The data items on which the operators act upon are called operands. Some operators require two operands while others act upon only one operand. They are classified as:

1. Arithmetic Operators (binary type)
2. Unary Operators
3. Relational and Logical Operators
4. Assignment Operator

Arithmetic Operators:

In C, most programs perform arithmetic calculations. Arithmetic calculations can be performed by using the following arithmetic operators. Table 8.1 summarizes the C arithmetic operators. Note the use of various special symbols not used in algebra. The asterisk (*) indicates multiplication and the percent sign (%) is the modulus or remainder operator. The arithmetic operators in the Table are all binary operators, i.e., operators that take two operands.

C++ operation	C++ arithmetic operator	C++ expression
Addition	+	$x + y$
Subtraction	-	$x - y$
Multiplication	*	$x * y$
Division	/	x / y
Modulus	%	$x \% y$

Table8.1:Arithmeticoperators

UnaryOperators:

In addition to the arithmetic assignment operators, C++ also provides two unary operators that act upon a single operand to produce a new value, for adding 1 to or subtracting 1 from the value of a numeric variable. These are the unary increment operator, `++`, and the unary decrement operator, `--`, which are summarized in the Table 8.2.

Operators	Operation	Explanation
<code>++a</code>	Pre Increment	Increment <i>a</i> by 1, then use the new value of <i>a</i> in the expression in which <i>a</i> resides.
<code>a++</code>	Post Increment	Use the current value of <i>a</i> in the expression in which <i>a</i> resides, then increment <i>a</i> by 1.
<code>--a</code>	Pre Decrement	Decrement <i>a</i> by 1, then use the new value of <i>a</i> in the expression in which <i>a</i> resides.
<code>a--</code>	Post Decrement	Use the current value of <i>a</i> in the expression in which <i>a</i> resides, then decrement <i>a</i> by 1.

Table8.2:Incrementanddecrementoperators

AssignmentOperators:

C++ provides several assignment operators for abbreviating assignment expressions. For example, the statement: `c = c + 3;`

The `+=` operator adds the value of the expression on the right of the operator to the value of the variable on the left of the operator and stores the result in the variable on the left of the operator. Thus the assignment `c += 3` adds 3 to *c*. Table 8.3 shows the arithmetic assignment operators, sample expressions using these operators and explanations.

Assignment operator	Sample expression	Explanation	Assigns
<i>Assume: int c = 3, d = 5, e = 4, f = 6, g = 12;</i>			
Subtraction -=	d -= 4	d = d - 4	1 to d
Multiplication *=	e *= 5	e = e * 5	20 to e
Division /=	f /= 3	f = f / 3	2 to f
Remainder %=	g %= 9	g = g % 9	3 to g

Table8.3:Arithmeticassignmentoperators

SummaryofOperatorPrecedenceandAssociativity:

Table8.4addsthe logicaloperatorstotheoperatorprecedenceandassociativitychart. Theoperatorsareshownfromtoptobottom, indecreasingorderofprecedence.

LAB TASK:

Task 1: Write a program which take input from user & perform arithmetic operations i.e. calculations.

Task 2: Write a program which will convert Fahrenheit into Centigrade and Kelvin.

Task 3: Calculate your age in Months, Weeks, Days, Hours, and Minutes etc.

Date: _____

Lab#11DECISIONMAKINGINCPROGRAMMING

OBJECTIVES:

- O1: TounderstandandhowtousedecisionmakingstatementinClanguage.
- O2: Createdifferentprogramsrelatedtodecisionmakingstatement.

EQUIPMENTTOBEUSEDINTHELAB

- Acomputersystem
- DEVCppinstalled

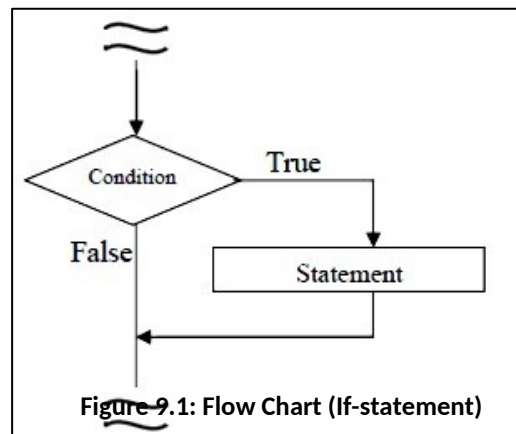
THEORY:

Normally, your program flows along line by line in the order in which it appears in your source code. But, it is sometimes required to execute a particular portion of code only if certain condition is true; or false i.e. you have to make a decision in your program. There are three major decision-making structures. Four decision-making structures:

1. If statement
2. If-else statement
3. Switch case
4. Conditional Operator (Rarely used)

The if statement:

The if statement enables you to test for a condition



(such as whether two variables are equal) and branch to different parts of your code, depending on the result. The simplest form of an if statement is:

```
if(expression)
```

```
statement;
```

The expression may consist of logical or relational operators like (>>=<=&&||)

The if-else statement:

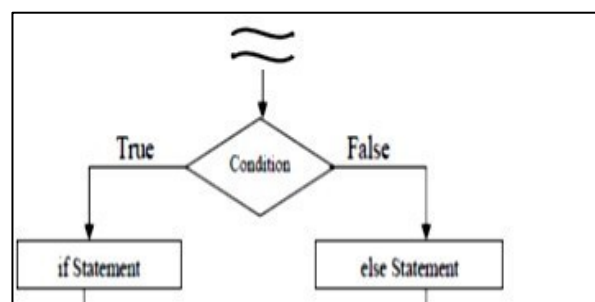




Figure 9.1: Flowchart (If-else statement)

Often your program will want to take one branch if your condition is true, another if it is false. The keyword `else` can be used to perform this functionality:

```
if(expression) statement; else statement;
```

Note: To execute multiple statements when a condition is true or false, parentheses are used.

Consider the following example that checks whether the input character is an uppercase or lowercase:

LAB TASK:

1. Write a program that takes a number as input from user and checks whether the number is even or odd.

- a) Using `if-else`
- b) Using conditional operator

2. Write a program that declares and initializes two numbers with `your_roll_no` and `your_friend_roll_no` and displays the greater of the two.

Date: _____

Lab#12.LOOPING CONSTRUCTS IN C LANGUAGE.

OBJECTIVES:

- O1: To understand the different loops constructs in C language.
- O2: To use loops in different programs.

EQUIPMENT TO BE USED IN THE LAB

- A computer system
- DEV C++ installed

Theory:

The concept of looping provides a way to execute a set of instructions more than once until a particular condition is true. In C, loop is constructed by three ways.

Types of loops:

1) for Loop

- i. simple for loop ii. Nested for loop

2) while Loop

- i. simple while loop ii. Nested while loop

3) do-while Loop

- i. simple do while loop ii. Nested do while loop

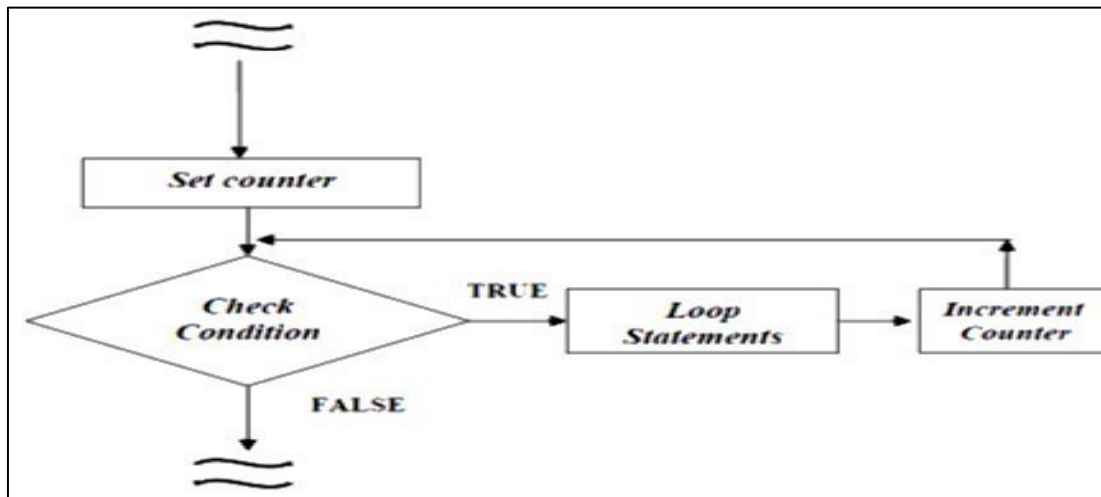
The for Statement

The for loop is appropriate when you know in advance how many times the loop will be executed. Here you have a counter variable whose limits are defined. The general form of the for statement is

for(initialization of counter; loop continuation condition; increment counter)

Statements;

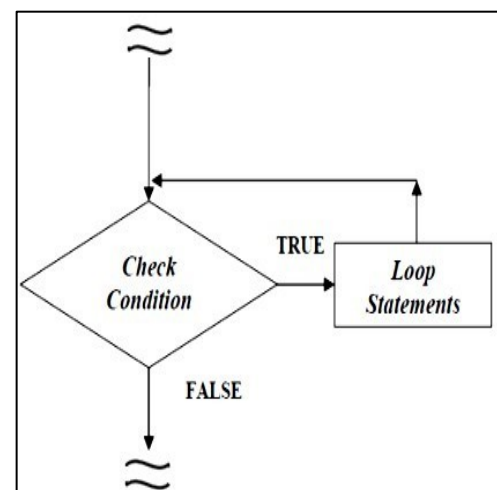
The initialization expression initializes the loop's control variable or counter (it is normally set to 0); loop continuation condition determines whether the loop counter is within its limits or not and finally the increment statement increments the counter variable after executing the inner loop statements. The flowchart of the for loop can be shown as



e10.1:FlowChart(forloop)

Figur

The while Statement:



The while loop is used to carry out looping operations, in which a group of statements is executed repeatedly, if condition following while is true otherwise control is transferred to the end of the loop. Here we do not know how many times the loop will be executed.

The general form of the while statement is while(condition)

{statement1;

:

statement2;}

The do while Statement

Fig10.3:Flowchart(While Loop)

The do while repetition statement is similar to the while statement. In the while statement, the loop-continuation condition test occurs at the beginning of the loop before the body of the loop executes. The do while statement tests the loop-continuation condition after the loop body executes; therefore, the loop body always executes at least once.

LAB TASKS:

1. Write a program to generate a series of first 50 even numbers
2. Write a program to generate tables from 2 to 20 with first 10 terms
3. Write two program segments, which may be used to input a sentence. Terminate when Enter key is pressed. (Use for and while loops).
4. Write a program to enter the number still the user wants and at the end it should display the count of positive, negative and zero entered.
5. Write a program to find the range of a set of numbers. Range is the difference between the smallest and biggest number in the list.
6. Write a program to display the following patterns.

<pre>* *</pre>	<pre>1 1 2 1 1 2 3 2 1 1 2 3 4 3 2 1 1 2 3 4 5 4 3 2 1</pre>
--	--

Date: _____

Lab#12DECISIONMAKINGTHESWITCHCASEANDCONDITIONALOPERATOR

OBJECTIVES:

- O2: To understand the Case Control Structure in C language.
- O2: How these control structures are used in preliminary programs..

EQUIPMENT TO BE USED IN THE LAB

- A computer system
- DEV C++ installed

THEORY:

Decisions using switch.

Normally, your program flows along line by line in the order in which it appears in your source code. But, it is sometimes required to execute a particular portion of code only if a certain condition is true; or false i.e. you have to make a decision in your program. There are three major decision-making structures. The 'if' statement, the if-else statement, and the switch statement. Another less commonly used structure is the conditional operator.

The switch Statement

Unlike if, which evaluates one value, switch statements allow you to branch on any of a number of different values. There must be a break at the end of the statements of each case; otherwise all the preceding cases will be executed including the default condition. The general form of the switch statement is:

```
switch(identifier variable)
{
    case identifierOne: statement;
                                break;
    case identifierTwo: statement;
                                break;
    ....
    case identifierN: statement;
                                break;
    default: statement;
}
```

The control statement that allows you to make a decision from the number of choices is called a switch, or more correctly as switch-case-default, since these three keywords go together to make up the control statement. They most often appear as follows:

```

switch(integerexpression)
{
caseconstant1:
dothis;

caseconstant2:
dothis;

caseconstant3:
dothis;

default:
dothis;
}

```

The integer expression following the keyword **switch** is any C expression that will yield an integer value. It could be an integer constant like 1, 2 or 3, or an expression that evaluates to an integer. The keyword **case** is followed by a **integer** or a **character Constant**. Each constant in each case must be different from all the others. The “do this” line in the above form of **switch** represents any valid C statement.

What happens when we run a program containing a **switch**?

First, the integer expression following the keyword **switch** is evaluated. The value it gives is then matched, one by one, against the constant values that follow the case statements. When a match is found, the program executes the statements following that case, and all subsequent case and default statements as well. If no match is found with any of the case statements, only the statements following the default are executed. A few examples will show how this control structure works.

Even if there are multiple statements to be executed in each case there is no need to enclose them within a pair of braces (unlike **if**, and **else**).

Every statement in a **switch** must belong to some case or the other. If a statement doesn't belong to any case the compiler won't report an error. However, the statement would never get executed. For example, in the following program the **printf()** never goes to work.

switch Versus if-else Ladder

There are some things that you simply cannot do with a **switch**. These are:

- (a) A float expression cannot be tested using a **switch**
- (b) Cases can never have variable expressions (for example it is wrong to say **case a+3:**)

(c) Multiple cases cannot use same expressions.

(a),

(b) and (c) above may lead you to believe that these are obvious disadvantages with a **switch**, especially since there weren't any such limitations with **if-else**.

Lab Tasks:

- Write a program which takes a text input, counts total number of vowels, consonants and other special characters and prints the result.
- Write a program to make a simple calculator which should be able to do +, -, *, /, % operations
- Write a program which takes 10 integers as input and prints the largest one.

Learning Outcomes:

- 1- The students should be able to use Decisions Using switch.
- 2- Students must be capable to implement control statements effectively.

Open ended II